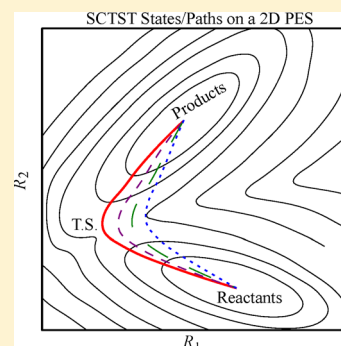


New Developments in Semiclassical Transition-State Theory

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ABSTRACT: This Feature Article describes some recent developments and applications of the Semiclassical Transition-State Theory (SCTST) for treating quantum tunneling in chemical reactions. A reduced dimensional form of the SCTST is discussed and is shown to be particularly efficient, as the required number of electronic structure calculations is reduced to an absolute minimum. We also describe how an alternative formulation of SCTST developed by Hernandez and Miller [*Chem. Phys. Lett.* **1993**, 214, 129], the SCTST- θ , has advantages in allowing for straightforward applications of the SCTST for any form of the potential expansion at the transition state. We also illustrate the power of SCTST in applications to more complex systems. We show how polyatomic modes such as internal rotations and torsions can be treated efficiently in SCTST calculations. We also describe some applications of the method to hydrogen atom tunneling in unimolecular reactions including the degradation of chemical nerve agents and the decay of the atmospherically important Criegee intermediates.



1. INTRODUCTION

The development of efficient and reliable methods for calculating rate constants of chemical reactions is a very active area of theoretical chemistry.^{1,2} Transition-state theory is very widely used and can be applied directly and automatically from modern electronic structure computer packages.³ However, this theory in its simplest form neglects quantum tunneling effects, and these can be significant for reactions involving the abstraction or exchange of hydrogen atoms. Therefore, developing reaction rate theories that treat tunneling continues to receive much attention.⁴

The most complete theory that incorporates tunneling is quantum reactive scattering (QRS) in full dimensions using an accurate potential energy surface (PES) obtained from ab initio quantum chemistry calculations.^{5,6} This theory can be applied using time-dependent or time-independent quantum dynamics. However, the computational cost rises exponentially with the increase of dimensionality, and this currently limits the general applicability of the full approach to reactions involving six atoms or fewer. Such calculations, however, do serve as a very useful benchmark to test the accuracy of more approximate theories.⁵ Theories based on transition-state theory with tunneling corrections have been very effectively applied to many reaction problems.^{3,7} However, the accuracy of such approaches can in some cases be uncertain. Other promising methods include Ring Polymer Molecular Dynamics (RPMD),⁸ which exploits path integrals and involves a relatively minor adaptation of commonly used molecular dynamics codes to treat quantum effects, and the Instanton approach, which enables the calculation of tunneling through a single periodic orbit in the reaction configuration space.⁹

Since chemical reaction rates depend exponentially on the activation energy through the Arrhenius relation, an electronic structure calculation of the highest accuracy is required to

compute the height of reaction barriers.¹⁰ This means that, despite considerable recent progress,¹¹ few global PESs have the required accuracy for reactions of larger polyatomic molecules. This limits the general applicability of most of the theories described above for treating quantum tunneling.

Semiclassical Transition-State Theory (SCTST) is a computationally inexpensive method for calculating rate constants, which includes the treatment of tunneling.^{12–14} The method has the distinct advantage that a global PES in all degrees of freedom (DOF) from reactants to products is not needed. Instead all that is required is an expansion of the force field around the transition state. The tunneling probabilities are obtained from simple analytical formulas in the harmonic and anharmonic force constants, which can be obtained directly from ab initio electronic structure calculations.¹⁴ This theory was developed by Miller and co-workers in the 1970s,^{12,13} but its accuracy was not clear at that time due to the small number of accurate benchmark quantum dynamics calculations that were available. It has become clear more recently that this theory can give very reliable results for quantum tunneling,^{14–17} and hence SCTST has emerged as a very powerful and direct approach for computing rate constants.

Reduced dimensionality (RD) methods make quantum dynamics calculations more cost-effective and applicable.¹⁸ A general form of RD theory that we developed and tested involves treating the chemical bonds being broken or formed using explicit quantum dynamics, while the effect of the spectator vibrational modes are considered by including their harmonic frequencies in the PES.^{19–23} This combined reduced

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dimensional and quantum chemistry (RDQC) approach has been applied to many reactions of polyatomic molecules.⁴ It requires only a minimal number of accurate quantum chemistry computations, and the quantum dynamics calculations are very inexpensive to perform. The approach has been used to test the accuracy of the SCTST for the same PESs for a range of hydrogen atom transfer reactions of polyatomic molecules, and the agreement between the two sets of results is excellent in every case.^{4,16,17} Combining the RD approach with SCTST also produces a highly efficient theory that will be applicable to a very wide range of reactions of larger polyatomic molecules.²⁴

In this Feature Article we describe recent developments in the SCTST with particular emphasis on its application within the RD approach. Computations on a range of bimolecular and unimolecular chemical reactions are described. A short theoretical background overview of the original form of the SCTST, and examples of its applications to chemical reactions in both full dimensions (FD) and one dimension (1D), is given in Section 2. In Section 3, we consider the SCTST in the barrier penetration integral domain, which introduces some novel features to the approach. In addition, implementation of SCTST with deep tunneling corrections and fourth-order vibrational perturbation theory within the RD framework is also included for both bimolecular and unimolecular reactions. In Section 4, we discuss the latest applications and developments of the RD-SCTST including corrections to the 1D SCTST and the applications to corner-cutting heavy-light-heavy systems. Conclusions are given in Section 5, where we emphasize that SCTST has considerable potential for the efficient calculation of reliable rate constants for a wide variety of chemical reactions.

2. THEORETICAL BACKGROUND

2.1. TST. In a standard rigid rotor harmonic oscillator (RRHO) transition-state theory (TST) calculation, the rate constant is given by

$$k_{\text{TST}}(T) = \frac{1}{h\beta} \frac{Q^\ddagger(T)}{Q^{\text{R}}(T)} \exp(-\beta\Delta V_f^\circ) \quad (1)$$

where $\beta = k_{\text{B}}^{-1}T^{-1}$, ΔV_f° is the adiabatic reaction barrier in the forward direction, and $Q^\ddagger(T)$ and $Q^{\text{R}}(T)$ are the total partition functions of the transition state and reactant, respectively. They are given by

$$Q(T) = Q_{\text{elec}}(T)Q_{\text{trans}}(T)Q_{\text{rot,RR}}(T)Q_{\text{vib,HO}}(T)$$

It is also possible to compute the rate constant using more accurate anharmonic vibrational partition functions. Equation 1 can then be written as

$$k_{\text{TST,anh}}(T) = \frac{1}{h\beta} \frac{Q_{\text{anh}}^\ddagger(T)}{Q_{\text{anh}}^{\text{R}}(T)} \exp(-\beta\Delta V_f^\circ) \quad (2)$$

Note that, for most reactions, this error introduced by the harmonic oscillator model is negligible because of the cancellation of errors when calculating the ratio between the TS and reactant partition functions, $Q_{\text{vib}}^\ddagger/Q_{\text{vib}}^{\text{R}}$. To achieve this cancellation of errors, one needs to use the same vibrational model for the reactant and TS. In other words, we have

$$\frac{Q_{\text{vib,anh}}^\ddagger}{Q_{\text{vib,anh}}^{\text{R}}} \simeq \frac{Q_{\text{vib,HO}}^\ddagger}{Q_{\text{vib,HO}}^{\text{R}}} \neq \frac{Q_{\text{vib,anh}}^\ddagger}{Q_{\text{vib,HO}}^{\text{R}}} \text{ or } \frac{Q_{\text{vib,HO}}^\ddagger}{Q_{\text{vib,anh}}^{\text{R}}} \quad (3)$$

It is then easy to see that $k_{\text{TST}}(T) \simeq k_{\text{TST,anh}}(T)$. In addition, it is often very expensive to conduct anharmonic analyses for polyatomic molecules; thus, the most commonly used TST rate constant calculation follows eq 1.

A common misconception is that all SCTST models are improvements to the standard TST given in eq 1. In reality, however, only 1D SCTST is based on eq 1, while the FD SCTST is based on the anharmonic TST in eq 2. The multidimensional or RD SCTST lies somewhere between the two. Details of the treatment and challenges of RD SCTST can be found in Section 4. Here we discuss the FD SCTST and 1D SCTST.

2.2. FD SCTST. In an FD SCTST calculation, the reaction rate constant is given by^{12–14,25–27}

$$k_{\text{SCTST}}(T) = \frac{1}{h} \frac{Q^{\ddagger'}(T)}{Q_{\text{anh}}^{\text{R}}(T)} \int_{E_{\text{thresh}}}^{\infty} N_{\text{FD}}(E_{\text{v}}) \exp(-\beta E_{\text{v}}) dE_{\text{v}} \quad (4)$$

where $Q^{\ddagger'}(T)$ is the partition function of the TS consisting of the electronic, rotational, and translational partition function. Here, $N_{\text{FD}}(E_{\text{v}})$ is the cumulative reaction probability (CRP). It is calculated by summing over the TS vibrational configuration dependent reaction probability.

$$N_{\text{FD}}(E_{\text{v}}) = \sum_{\{n^\ddagger\}} P_{\{n^\ddagger\}}(E_{\text{v}}) \quad (5)$$

Here, $\{n^\ddagger\}$ corresponds to the set of quantum numbers of the nonreactive (bound) vibrational states of the TS. Each unique set is referred to as a TS vibrational configuration. Combining these two equations, we have

$$k_{\text{SCTST}}(T) = \frac{1}{h} \frac{Q^{\ddagger'}(T)}{Q_{\text{anh}}^{\text{R}}(T)} \sum_{\{n^\ddagger\}} \int_{E_{\text{thresh}}}^{\infty} P_{\{n^\ddagger\}}(E_{\text{v}}) \times \exp(-\beta E_{\text{v}}) dE_{\text{v}} \quad (6)$$

Also note that, if quantum tunneling effects were not included in the computation of $P_{\{n^\ddagger\}}$, then eq 6 would reduce to eq 2. In SCTST, the semiclassical formula is used to calculate $P_{\{n^\ddagger\}}$. It is given by^{12,13,25,26}

$$P_{\{n^\ddagger\}}(E_{\text{v}}) = \{1 + \exp[2\theta_{\{n^\ddagger\}}(E_{\text{v}})]\}^{-1} \quad (7)$$

where θ is the barrier penetration integral.

We now consider a TS that consists of F vibrational modes, where the F th mode is the reaction mode with an imaginary frequency. In the SCTST developed by Miller et al.,¹² to determine an analytical expression for θ in terms of easily calculable quantities, the Bohr–Sommerfeld theory is used. The reaction mode, or F th action variable, is given by

$$2\pi\hbar(n_F + 1/2) = \oint p_F dq_F = 2 \int p_F dq_F \quad (8)$$

where p_F is the momentum conjugate to position q_F . For a particular TS vibrational configuration $\{n^\ddagger\}$, it can be written as

$$p_F = [E_{\text{v}} - V_{\{n^\ddagger\}}(q_F)]^{1/2} \quad (9)$$

Combining these two equations, we get

$$\frac{\pi(n_F + 1/2)}{i} = \frac{1}{\hbar} \int [V_{\{n^\ddagger\}}(q_F) - E_v]^{1/2} dq_F \quad (10)$$

where the right side of this equation is equivalent to a barrier penetration integral, $\theta_{\{n^\ddagger\}}$.^{12,13} We thus have

$$\frac{(n_F + 1/2)}{i} = \frac{\theta_{\{n^\ddagger\}}(E_v)}{\pi} \quad (11)$$

Perturbation theory can then be used to connect $(n_F + 1/2)$ to the energy of an incoming wavepacket. Second-order vibrational perturbation theory (VPT2)²⁶ gives the potential energy at the TS as

$$\begin{aligned} E_v(n^\ddagger, \theta) = & \Delta V_f + G_0 + \sum_{j=1}^{F-1} \hbar \omega_j (n_j + 1/2) \\ & + \sum_{j=1}^{F-1} \sum_{k=j}^{F-1} x_{jk} (n_j + 1/2) (n_k + 1/2) \\ & - \left[\frac{\hbar \omega_F}{i} + \sum_{j=1}^{F-1} \frac{x_{jF}}{i} (n_j + 1/2) \right] \\ & \times \left[\frac{(n_F + 1/2)}{i} \right] - x_{FF} \left[\frac{(n_F + 1/2)}{i} \right]^2 \end{aligned} \quad (12)$$

Substituting eq 11 into eq 12 and solving for θ gives

$$\begin{aligned} \theta_{\{n^\ddagger\}}(E_v) = & \frac{\pi(-\Omega_{\{n^\ddagger\}} + [\Omega_{\{n^\ddagger\}}^2 + 4x_{FF}(\Delta V_{f(\text{anh}),\{n^\ddagger\}} - E_v)]^{1/2})}{2x_{FF}} \end{aligned} \quad (13)$$

where

$$\Omega_{\{n^\ddagger\}} = \frac{\hbar \omega_F}{i} + \sum_{i=1}^{F-1} \frac{x_{iF}}{i} (n_i + 1/2) \quad (14)$$

$$\begin{aligned} \Delta V_{f(\text{anh}),\{n^\ddagger\}} = & \Delta V_f + G_0 + \sum_{j=1}^{F-1} \hbar \omega_j (n_j + 1/2) \\ & + \sum_{j=1}^{F-1} \sum_{k=j}^{F-1} x_{jk} (n_j + 1/2) (n_k + 1/2) \end{aligned} \quad (15)$$

where the ω_F is the reaction mode vibrational frequency and x_{jk} are the anharmonic constants. Note that, in eq 15, the forward reaction barrier ΔV_f does not include the zero point energy (ZPE) corrections and is different from ΔV_f° in eqs 1 and 2. The third- and fourth-order derivatives of the PES with respect to the vibrational normal modes are required to calculate x_{jk} using VPT2.^{14,26,27,31–33} Detailed VPT2 derivations can be found in refs 31, 26, 32. These higher-order derivatives can be obtained from an analytically fitted PES^{17,24} or from ab initio calculations.^{14,15,27,34}

We use the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction to illustrate the application of SCTST to a real chemical reaction. The results are shown in Figure 1. In Figure 1a, the higher-order derivatives were obtained from a two-dimensional analytical PES.^{17,28} The analytical PES function was fitted to ab initio energies calculated at CCSD(T)//MP2/cc-pVTZ level for grid points along two active vibrational modes.²⁸ The SCTST result (black curve) shows good agreement compared to the rate

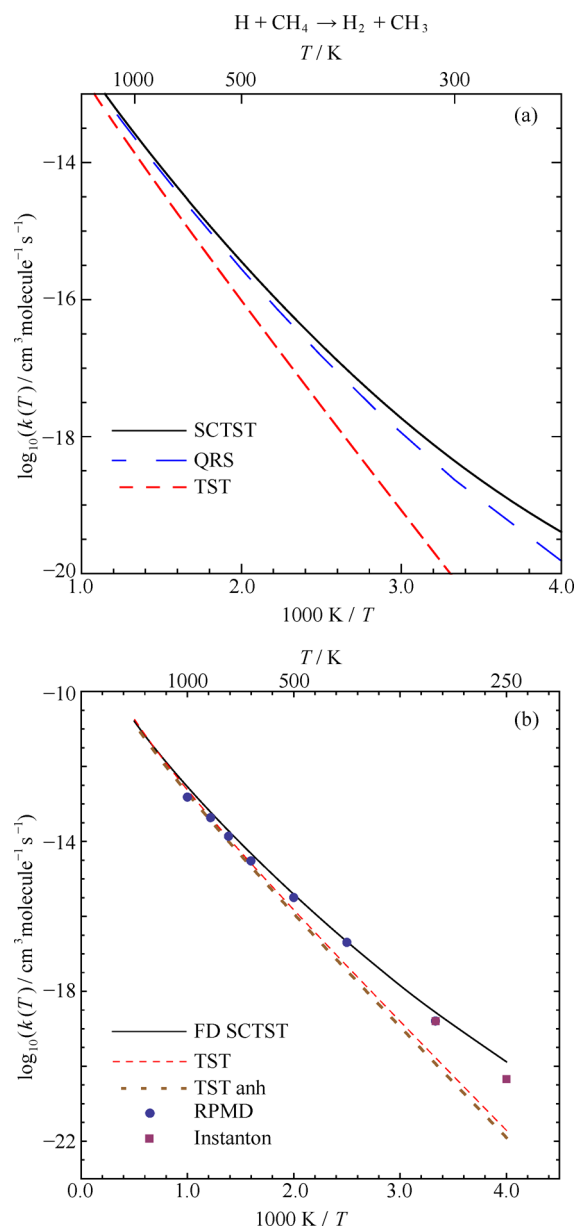


Figure 1. Application of SCTST to the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction. (a) Arrhenius plot of the rate constants calculated using SCTST based on 2D PES,²⁸ comparing to the QRS results based on the same surface. (b) Arrhenius plot of the rate constants calculated using FD SCTST, standard TST, and anharmonic TST, comparing to RPMD²⁹ and Instanton³⁰ results.

constant from a QRS calculation (dashed blue curve) performed on the same PES.

In Figure 1b, the SCTST results were calculated directly from ab initio calculations at CCSD(T)//MP2/aug-cc-pVTZ level. Comparing the standard TST (dashed red curve) and the anharmonic TST (dashed brown curve), we can see the relationship given in eq 3 stands. The FD SCTST rate constant converges to the TST results at high temperature. The quantum tunneling contribution becomes more pronounced at temperatures typically lower than 500 K. We can also see that FD SCTST has an excellent agreement when compared to other theoretical results^{29,30} at these temperatures.

2.3. 1D SCTST. The FD SCTST has two major sources of computational costs. The computation of the third- and fourth-

order derivatives of the PES can quickly become exceedingly expensive, as the size of system increases.^{14,27,33} In addition, the calculation of the cumulative reaction probability via eq 5 is also a formidable task, because the number of TS configurations, $\{n^\ddagger\}$, increases exponentially with the number of DOF. Enumeration of all possible configurations for even 3 or 4 degrees of freedom is very computationally expensive.³⁴ For larger systems, one can use the Wang–Landau algorithm^{35–37} and its implementation for parallel architectures³⁸ to obtain an approximate density of states and tunneling probability. Generally speaking, even with these methods, the FD SCTST can quickly become too expensive when studying relatively large systems. It is therefore desirable to develop an approximate method based on the principle of SCTST but scales better with system size. We first discuss here the one-dimensional approach, which has been applied to various reactions, including H abstractions,^{24,33,39} H exchange,⁴⁰ and unimolecular H transfers.^{41,42} In Section 4, we shall explain the limitation of this method and suggest possible solutions.

To simplify the FD SCTST expression of rate constants in eq 6 into the 1D SCTST equation, three approximations are made. First, the reaction mode is assumed to be decoupled from the other bound vibrational modes. This allows the separation of the summation in the CRP calculation from the integration over energy in the rate constant calculation. We have

$$k(T) = \frac{1}{h} \frac{Q_{\text{anh}}^\ddagger(T)}{Q_{\text{anh}}^R(T)} \int_{E_{\text{thresh}}}^\infty P_{\{0\}}(E_v) \exp(-\beta E_v) dE_v \quad (16)$$

Here the CRP is replaced by the ground-state reaction probability $P_{\{0\}}(E_v)$, and the total partition function of the TS $Q_{\text{anh}}^\ddagger(T)$ now includes the anharmonic vibrational partition functions for all the bound modes.

The second assumption utilizes the relationship given in eq 3 that we assume all the bound vibrational modes in both TS and reactants can be treated harmonically. The rate constants equation can then be written as

$$k(T) = \frac{1}{h} \frac{Q^\ddagger(T)}{Q^R(T)} \int_{E_{\text{thresh}}}^\infty P_{\{0\}}(E_v) \exp(-\beta E_v) dE_v \quad (17)$$

Note that, in the calculation of $P_{\{0\}}$, the harmonic ground-state energies are used. Following eq 13 and the approximations given so far, x_{FF} and G_0 are the two required parameters to compute this rate constant. In an FD VPT2 calculation, they are given by

$$x_{\text{FF}} = \frac{\hbar}{16\omega_F^2} \left(f_{\text{FFFF}} - \sum_{i=1}^F \frac{f_{\text{FF}i}^2}{\omega_i^2} \frac{8\omega_F^2 - 3\omega_i^2}{4\omega_F^2 - \omega_i^2} \right) \quad (18)$$

and

$$G_0 = \frac{\hbar}{64} \frac{f_{\text{FFFF}}}{\omega_F^2} - \frac{7\hbar^2}{576} \frac{f_{\text{FFF}}^2}{\omega_F^4} + \frac{3\hbar^2}{64} \sum_{i \neq F} \frac{f_{iF}^2}{(4\omega_i^2 - \omega_F^2)\omega_i^2} - \frac{\hbar^2}{4} \sum_{i < j < F} \frac{f_{ijF}^2}{[(\omega_i + \omega_F)^2 - \omega_j^2][(\omega_i - \omega_F)^2 - \omega_j^2]} \quad (19)$$

where f_{FFF} and f_{FFFF} are the third- and fourth-order derivatives of the PES with respect to the reaction mode, respectively. f_{iF}

and f_{ijF} are the coupling third-order derivatives between the reaction mode and bound modes. The computation of these derivatives using ab initio calculations can still be considerably expensive.

To further reduce the computational cost of the calculation, in the final approximation of 1D SCTST, we assume that all coupling terms in the third-order derivatives are 0 between the reaction mode and the bound modes. Therefore, in 1D SCTST, we have

$$x_{\text{FF}}^{\text{1D}} = \frac{\hbar}{16\omega_F^2} \left(f_{\text{FFFF}} - \frac{5f_{\text{FFF}}^2}{3\omega_F^4} \right) \quad (20)$$

and

$$G_0^{\text{1D}} = \frac{\hbar}{64} \frac{f_{\text{FFFF}}}{\omega_F^2} - \frac{7\hbar^2}{576} \frac{f_{\text{FFF}}^2}{\omega_F^4} \quad (21)$$

Now, it is clear to see that all the additional cost of a 1D SCTST to a standard TST is the computation of f_{FFF} and f_{FFFF} , which can be easily done using numerical differentiation of single-point energies at geometries displaced along the reaction mode. At a minimum, only four single-point energies are required, in addition to a standard TST calculation.

We show, in Figure 2, the rate constants for the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction calculated using the different level of

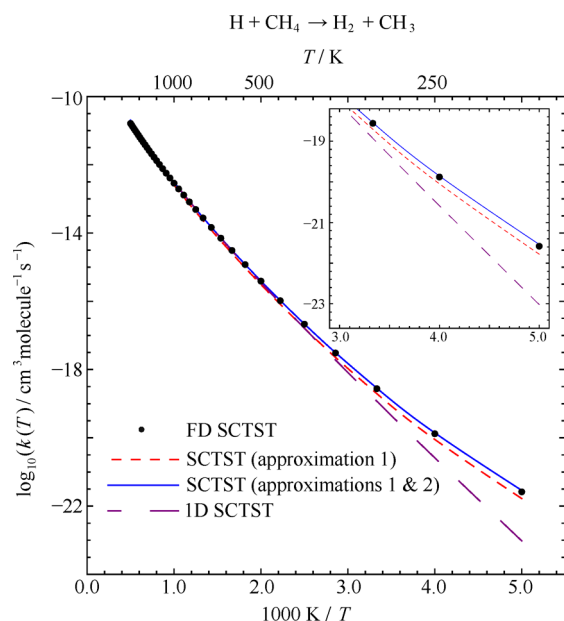


Figure 2. Application of FD SCTST and approximated SCTST to the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction.

approximations discussed here. It can be seen that the rate constants calculated after the first two levels of approximations show very good agreement to the FD SCTST results. The simplest form of 1D SCTST agrees well with the FD SCTST for temperatures above 300 K, but it underestimates the rate as temperature decreases.

2.4. Deep Tunneling. In reaction rate constant studies, deep tunneling (DT) refers to reactions proceeding at energies well below the barrier height. At 200 K, the simplest 1D SCTST typically underestimates the rate constant by ~ 30 times when compared to the FD SCTST. This underestimation is a result of the VPT2 expansion of the potential

around the TS being inaccurate in the reactant and product asymptotic regions. This problem has been discussed in great detail in ref 43. A solution given by Wagner et al.⁴³ is referred to as the DT correction to the SCTST in the current work. In the DT correction, the reaction potential is approximated by a piecewise continuous function comprised of three asymmetric Eckart segments chosen to match the forward and reverse barrier heights, the reaction mode frequency, and anharmonic constant. The segments are also chosen such that the barrier is continuous and smooth at all times. To apply the DT correction to the SCTST, the reverse barrier height of a reaction is also required.^{33,43}

We show in Figure 3a the 1D VPT2 barrier and the DT-corrected barrier for the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction; the three pieces of the DT-corrected barrier are marked with different colors. The rate constants with and without DT correction are shown in Figure 3b. The DT-corrected 1D SCTST has a much better agreement with the FD SCTST at

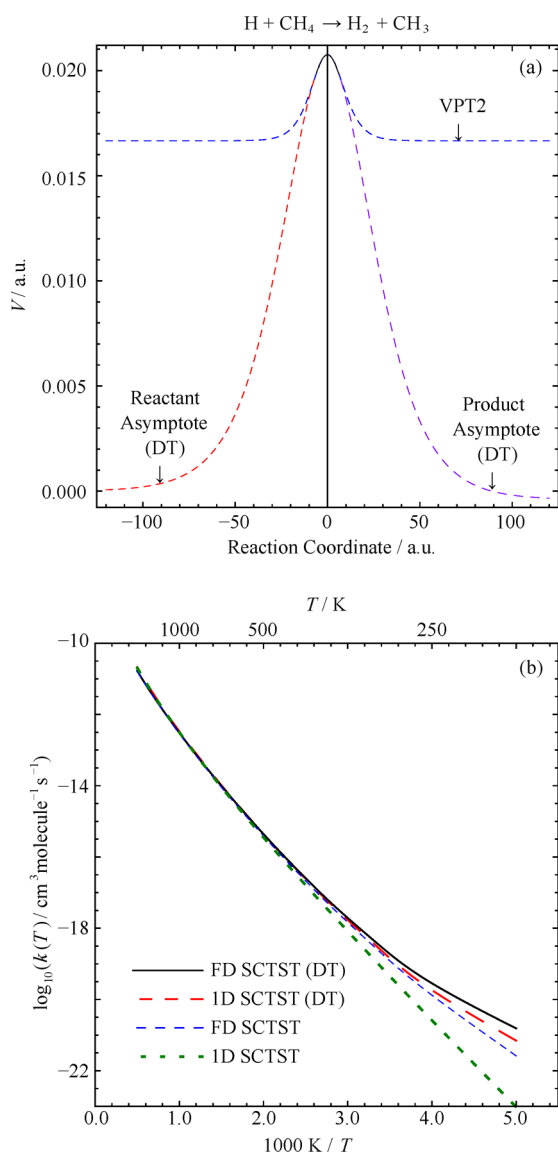


Figure 3. DT corrections to the SCTST calculations for the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction. (a) The comparison between DT corrected barrier and VPT2 barrier, (b) the comparison between the DT-corrected FD and 1D SCTST rate constants.

low temperatures. It is also interesting to note that the DT correction has a greater impact on the 1D calculation than the FD one.

3. DERIVATION OF SCTST- θ

3.1. General Derivation. In the previous section, we discussed the most common way of deriving SCTST as well as practical simplifications of the FD SCTST within the reduced dimensionality framework. However, this derivation can become complicated or even break down when a more accurate potential function is applied.^{43–45} For example, when a VPT4 potential is used rather than the VPT2, the relationship in eq 13 is obtained by solution of a cubic equation in θ ,^{44,45} and higher orders of perturbation theory would require analytical solutions to higher-order polynomials, which do not exist in general. In this section, we discuss an alternative formulation of SCTST proposed by Hernandez and Miller,⁴⁶ which we shall refer to as the SCTST- θ in this paper. This formulation allows for any potential $E(\theta)$ to be used and provides a more flexible and intuitive framework to view the SCTST.

In eq 6, the integration is performed in the energy domain. Hernandez and Miller⁴⁶ suggested that the same integration can also be done in the barrier penetration integral, θ , domain. The FD SCTST is then given by

$$k_{\text{FD SCTST-}\theta}(T) = \frac{1}{h} \frac{Q^\ddagger(T)}{Q_{\text{anh}}^{\text{R}}(T)} \times \sum_{\{n^\ddagger\}} \int_{\theta_{\text{T}}}^{-\infty} \frac{\exp[-\beta E_{\text{v}}(n^\ddagger, \theta)]}{1 + \exp(2\theta)} \frac{\partial E_{\text{v}}(n^\ddagger, \theta)}{\partial \theta} d\theta \quad (22)$$

where $E_{\text{v}}(n^\ddagger, \theta_{\text{T}}) = E_{\text{thresh}}$. Performing the integration by parts, we get

$$\begin{aligned} & \int_{\theta_{\text{T}}}^{-\infty} \frac{\exp[-\beta E_{\text{v}}(n^\ddagger, \theta)]}{1 + \exp(2\theta)} \frac{\partial E_{\text{v}}(n^\ddagger, \theta)}{\partial \theta} d\theta \\ &= \frac{\exp[-\beta E_{\text{v}}(n^\ddagger, \theta_{\text{T}})]}{\beta[1 + \exp(2\theta_{\text{T}})]} + \frac{1}{\beta} \\ & \quad \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \exp[-\beta E_{\text{v}}(n^\ddagger, \theta)] d\theta \\ &= \frac{\exp(-\beta E_{\text{thresh}})}{\beta[1 + \exp(2\theta_{\text{T}})]} + \frac{1}{\beta} \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \exp[-\beta E_{\text{v}}(n^\ddagger, \theta)] d\theta \end{aligned}$$

We note that the first term of the right-hand side of the equation is the Boltzmann weighted reaction probability at energy threshold E_{thresh} , which is negligible compared to the second part of the equation.⁴⁶ Equation 22 can be reduced to

$$k_{\text{FD SCTST-}\theta}(T) = \frac{1}{h\beta} \frac{Q^\ddagger(T)}{Q_{\text{anh}}^{\text{R}}(T)} \sum_{\{n^\ddagger\}} \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \times \exp[-\beta E_{\text{v}}(n^\ddagger, \theta)] d\theta \quad (23)$$

To use eq 23, it is only necessary to know the potential energy as a function of θ , and no inversion is required. As shown in eq 12, in VPT2, $E_{\text{v}}(n^\ddagger, \theta)$ is given by

$$E_{\text{v}}(n^\ddagger, \theta) = \Delta V_{\text{f(anh)}, \{n^\ddagger\}} - \Omega_{\{n^\ddagger\}} \left(\frac{\theta}{\pi} \right) - x_{\text{FF}} \left(\frac{\theta}{\pi} \right)^2 \quad (24)$$

Note that the FD SCTST rate constant given in eqs 23 and 24 is mathematically equivalent to the original formulation in eq 6, and hence it does not reduce the computational cost.

In this formulation, the 1D SCTST approximations become very easy to understand. The first two approximations simply remove the anharmonic terms apart from x_{FF} from eq 24; we then have

$$E_{\text{v}}^{\text{1D}}(n^{\ddagger}, \theta) = \Delta V_{\text{f}} + G_0 + \sum_{j=1}^{F-1} \hbar \omega_j (n_j + 1/2) - \frac{\hbar \omega_{\text{F}}}{i} \left(\frac{\theta}{\pi} \right) - x_{\text{FF}} \left(\frac{\theta}{\pi} \right)^2 \quad (25)$$

Substituting this into eq 23 and extracting all the bound vibrational modes of TS from the integration, we get

$$k_{\text{1D SCTST}-\theta}(T) = \frac{1}{h\beta} \frac{Q^{\ddagger}(T)}{Q^{\text{R}}(T)} \times \sum_{\{n^{\ddagger}\}} \exp \left\{ -\beta \left[\Delta V_{\text{f}} + \sum_{j=1}^{F-1} \hbar \omega_j (n_j + 1/2) \right] \right\} \times \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \times \exp \left[-\beta \left(G_0 - \frac{\hbar \omega_{\text{F}}}{i} \left(\frac{\theta}{\pi} \right) - x_{\text{FF}} \left(\frac{\theta}{\pi} \right)^2 \right) \right] d\theta \quad (26)$$

The summation term is now simply the vibrational partition function of the TS, so the 1D SCTST rate constant equation can be written as

$$k_{\text{1D SCTST}-\theta}(T) = k_{\text{TST}}(T) \kappa_{\text{1D}}(T) \quad (27)$$

where the tunneling correction $\kappa_{\text{1D}}(T)$ is given by

$$\kappa_{\text{1D}}(T) = \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \exp[-\beta E^{\text{anh(1D)}}(\theta)] d\theta \quad (28)$$

with

$$E^{\text{anh(1D)}}(\theta) = G_0 - \frac{\hbar \omega_{\text{F}}}{i} \left(\frac{\theta}{\pi} \right) - x_{\text{FF}} \left(\frac{\theta}{\pi} \right)^2$$

It is clear to see that, to carry out the 1D SCTST calculation, one only needs G_0 and x_{FF} , the acquisition of which was discussed in Section 2.2. Note that, although in eq 28 the tunneling is seemingly independent of the reaction barrier height, θ_{T} still must satisfy $E_{\text{v}}(n^{\ddagger}, \theta_{\text{T}}) = E_{\text{thresh}}$, for which the barrier height is required. The relationship between θ_{T} and E_{thresh} will be discussed in detail in Section 3.3.

3.2. Parabolic Limit. In the parabolic barrier limit ($x_{\text{FF}} = 0$), eq 28 becomes

$$\kappa_{\text{HO}}(T) = \int_{-\infty}^{\theta_{\text{T}}} \frac{\text{sech}^2(\theta)}{2} \exp \left[\frac{-i\beta \hbar \omega_{\text{F}} \theta}{\pi} \right] d\theta \quad (29)$$

This integration can be performed analytically:

$$\int \frac{\text{sech}^2(\theta)}{2} \exp \left[\frac{\tau \theta}{\pi} \right] d\theta = \frac{\exp \left(\frac{\tau \theta}{\pi} \right)}{2(\tau + 2\pi)} \times [-\tau \exp(2\theta) {}_2F_1 \left(1, 1 + \frac{\tau}{2\pi}, 2 + \frac{\tau}{2\pi}; -\exp(-2\theta) \right) + (\tau + 2\pi) {}_2F_1 \left(1, \frac{\tau}{2\pi}, 1 + \frac{\tau}{2\pi}; -\exp(2\theta) \right) + \tanh(\theta)] \quad (30)$$

where $\tau = -i\beta \hbar \omega_{\text{F}}$, and ${}_2F_1$ is the hypergeometric function. Note that this integration is performed as an indefinite integral. If the upper limit of the integration is chosen to be $+\infty$, then under the condition $0 \leq i\beta \hbar \omega_{\text{F}} < 2\pi$, eq 30 reproduces the Wigner tunneling correction⁴⁷

$$\kappa_{\text{HO}}(T)|_{-\infty}^{+\infty} = \kappa_{\text{W}}(T) = \frac{i\beta \hbar \omega_{\text{F}}/2}{\sin(i\beta \hbar \omega_{\text{F}}/2)} \quad (31)$$

When $i\beta \hbar \omega_{\text{F}} = 2\pi$, the corresponding temperature is often referred to as the crossover temperature T_{c} , at which the Wigner correction diverges.

Note, however, for most chemical reactions, θ_{T} in eq 29 is not $+\infty$. To calculate θ_{T} , one can simply solve a linear function of θ based on eq 24 with only the harmonic terms

$$\Delta V_{\text{f}}^{\circ} - \frac{\hbar \omega_{\text{F}}}{i\pi} \theta_{\text{T}} = E_{\text{thresh}}$$

For an exothermic reaction, $E_{\text{thresh}} = 0$, while for an endothermic reaction, $E_{\text{thresh}} = \Delta V_{\text{f}}^{\circ} - \Delta V_{\text{r}}^{\circ}$. Here $\Delta V_{\text{f}}^{\circ}$ is the adiabatic reverse barrier. Therefore, to compute the tunneling correction given in eq 29, one does not need any more ab initio calculations in addition to the standard TST.

A comparison between the calculated rate constants using the methods discussed so far in this work for the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction is shown in Figure 4. The reaction is exothermic, T_{c} is ~ 375.6 K, and θ_{T} for the modified Wigner rate is ~ 8.718 . We can see that, as expected, the Wigner rate constant diverges when approaching T_{c} . Although the modified Wigner result (dashed green curve) continues for T below T_{c} , it significantly overestimates the reaction rate. This is easily understood by comparing eq 29 to eq 28. x_{FF} is negative for this reaction, which results in a smaller area under curve for the integration for the 1D SCTST than the modified Wigner. Equivalently, a negative x_{FF} leads to a wider barrier than the parabolic one.

3.3. Deep Tunneling and VPT4 in the SCTST- θ . In Section 3.1, we derived the SCTST- θ proposed by Miller and co-workers⁴⁶ and showed how to implement the 1D SCTST in this framework. We now turn our focus to the key relationship, E_{v} as a function of θ . The VPT2 expression is given in eq 24. In Figure 5 we graphically show an example of this for the $\text{CH}_3 + \text{CH}_4$ reaction in the 1D approximation. VPT2 is based on a quadratic expansion of the surface about the TS, and hence it is expected to be accurate for $\theta \simeq 0$, since $\theta = 0$ corresponds to the TS. The upper integration limit of the SCTST- θ formula, θ_{T} , corresponds to the minimum value of E_{v} , E_{min} . For $\theta > \theta_{\text{T}}$, E_{v} is not physically meaningful. In principle, this minimum value of E_{v} should be equal to E_{thresh} given by

$$E_{\text{thresh}} = \begin{cases} 0 & \text{for symmetric or exothermic reactions} \\ \Delta V_{\text{f}} - \Delta V_{\text{r}} & \text{for endothermic reactions} \end{cases}$$

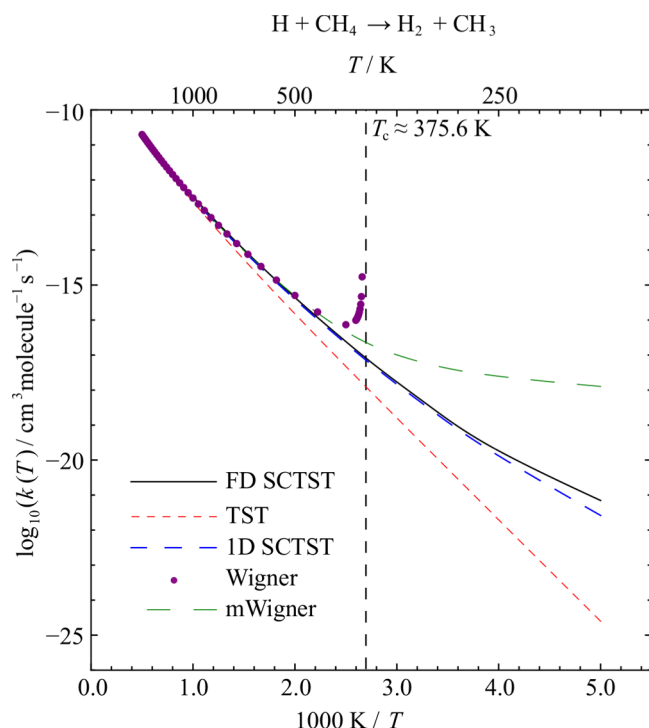


Figure 4. Comparison between Wigner, modified Wigner, 1D SCTST, FD SCTST, and standard TST rate constants for the $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$ reaction.

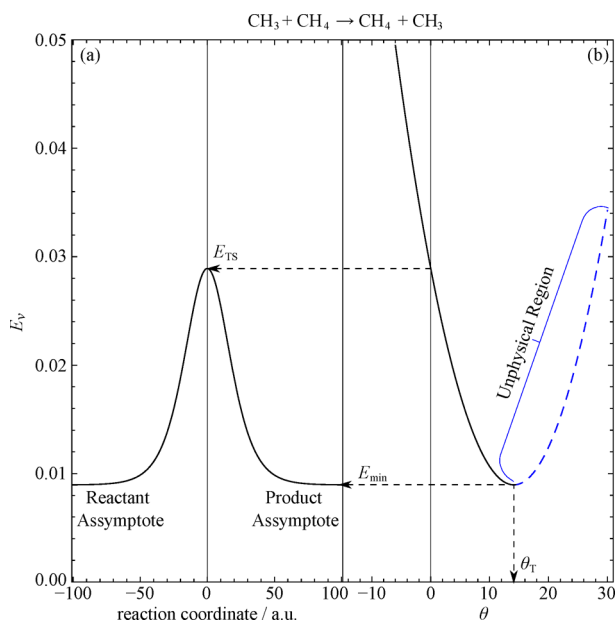


Figure 5. Plots of (a) the VPT2 barrier and (b) E_v as a function of θ for the $\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{CH}_3$ reaction.

where ΔV_r is the reverse barrier height. However, in the VPT2 calculation, $E_{\min}$ is often a poor approximation for E_{thresh} . In Figure 5, we have $E_{\min} > E_{\text{thresh}} = 0$. The VPT2 expression therefore ignores tunneling for $E_{\text{thresh}} < E < E_{\min}$ and thus underestimates the rate constants significantly at low T . This problem has been identified by Wagner et al.⁴³ Numerous works^{15,43–45} have been devoted to improving the SCTST in the deep tunneling regime. Here we discuss two methods and their implementation in the SCTST- θ formula.

The first method is to use a higher-order perturbation theory, for example, VPT4. Here the energy is given by

$$E(n^{\ddagger}, \theta) = V_0 + G - W \frac{\theta}{\pi} - X \left(\frac{\theta}{\pi} \right)^2 + Y \left(\frac{\theta}{\pi} \right)^3 \quad (32)$$

where V_0 is the $\{n^{\ddagger}\}$ state-dependent potential barrier height, G is the anharmonic correction to the potential, and W , X , and Y are given by analytical functions of up to sixth-order derivatives of the PES at the TS.⁴⁴ In the original formulation of SCTST, eq 32 would need to be solved for θ . This step is no longer necessary in SCTST- θ ; one can simply replace the VPT2 expression in eq 23 with eq 32. Moreover, it is also trivial to extend this to even higher-order perturbation theories. The VPT4 SCTST has been shown to improve behavior in the deep-tunneling regime for the collinear $\text{H} + \text{H}_2$ reaction⁴⁴ and several one-dimensional analytical potentials.⁴⁵ Clearly, however, this approach requires significantly more ab initio calculations, to determine the fifth- and sixth-order derivatives of the potential. Within the one-dimensional approximation this becomes tractable but, still, a significant expense.

A more efficient solution was determined by Wagner et al.⁴³ as discussed in Section 2.3. This correction can be much more easily implemented and understood in the SCTST- θ framework. In this case, the energy is effectively expanded to a fourth-order polynomial in θ , where the first two coefficients are as determined by VPT2, and the third- and fourth-order coefficients are chosen such that the forward and reverse barrier heights are correct and that the slope of $E(\theta)$ is zero at the reactant energy. An additional parameter $\theta_{\max}$ is required, which is equal to the barrier penetration integral at $E = 0$. This is calculated numerically using eq 33 and the composite potential discussed in ref. 43.

$$\theta_{\max} = \sqrt{2m} \int \sqrt{V(y)} dy \quad (33)$$

$E(\theta)$ is then given by

$$E(n^{\ddagger}, \theta) = \Delta V_r - \hbar \omega_{n^{\ddagger}}^* \frac{\theta}{\pi} - \hbar x_{\text{FF}} \left(\frac{\theta}{\pi} \right)^2 + \left[3\omega_{n^{\ddagger}}^* \left(\frac{\theta_{\max}}{\pi} \right) + 2x_{\text{FF}} \left(\frac{\theta_{\max}}{\pi} \right)^2 - 4\Delta V_r \right] \left(\frac{\theta}{\theta_{\max}} \right)^3 - \left[2\omega_{n^{\ddagger}}^* \left(\frac{\theta_{\max}}{\pi} \right) + x_{\text{FF}} \left(\frac{\theta_{\max}}{\pi} \right)^2 - 3\Delta V_r \right] \left(\frac{\theta}{\theta_{\max}} \right)^4 \quad (34)$$

If an exothermic reaction was studied, then ΔV_r in the coefficient of the third and fourth terms of θ should be replaced by ΔV_f . It is also interesting to note that this equation is akin to a VPT6 level expansion of energy, as it contains θ terms of the third and fourth orders. However, the coefficients of these terms in this equation are not derived from information at the TS, as one would have done in VPT6; instead, they are chosen specifically to correct the energy threshold problem.⁴³

3.3.1. Application of SCTST- θ . To illustrate the features of SCTST- θ discussed so far, we apply the various methods to the symmetric H-transfer reaction $\text{CH}_4 + \text{CH}_3$ in one dimension. The ab initio calculations were performed at CCSD(T)//MP2/aug-cc-pVTZ level of theory. The VPT4 coefficients were determined from derivatives obtained from a sixth-order polynomial function fitted to a total of 21 single-point energies

at the TS and the displaced geometries along the reaction coordinate. This method was able to reproduce previously calculated VPT2 coefficients⁴⁰ to good accuracy, as shown in Table 1.

Table 1. VPT2 and VPT4 Constants Calculated using a Fit to a Sixth-Order Polynomial, and As Calculated Previously using Richardson Extrapolation^a

	sixth-order polynomial	Richardson extrapolation ⁴⁰	difference
G	−81.645	−78.82	4%
X	−326.58	−315.36	4%
Y	−55.59		

^aValues in reciprocal centimeters.

Figure 6a shows how E_v varies as a function of θ for VPT2, VPT4 and Wagner's DT correction. Note that here, $E_{\text{thresh}} = 0$. Wagner's DT correction behaves correctly by construction, however both the VPT2 and the VPT4 fail to produce $E_{\text{min}} = 0$, the correct deep tunneling behavior.

The 1D SCTST- θ rate constants calculated using these three methods are shown in Figure 6b. By comparison to results obtained from a two-dimensional (2D) QRS calculation,²⁸ we can see that the DT-corrected VPT2 rate constant has the best agreement to the QRS for the entire temperature range. Both VPT2 and VPT4 underestimate the reaction rate, especially at temperatures below 300 K, where deep tunneling has a more significant contribution to the reaction. This result is in agreement with our prediction from the E_v versus θ plot in Figure 6a. It is also interesting to note that, for this particular reaction, the VPT4, having greater E_{min} than the VPT2, has a worse behavior in the deep tunneling regime than the VPT2.

In previous studies,^{44,45} VPT4 has been shown to improve behavior in the deep tunneling regime, in contrast to what we observe here. In these works, only one-dimensional potentials were studied.^{44,45} While our results^{44,45} suggest that VPT4 will improve the description of a one-dimensional, Eckart-like potential, this may not be true of a multidimensional, complex potential seen in real chemical systems. Accurately determining the threshold energy from derivatives at the TS requires substantial knowledge about the intermode couplings, since these couplings have a severe impact on the reaction path away from the TS. Thus, one expects only a very high order of perturbation theory could correctly describe the deep tunneling regime. It is thus clear that the method of Wagner⁴³ is a more efficient way to ensure the behavior is correct, without requiring detailed knowledge of these couplings. In fact a similar treatment can be implemented to the VPT4 calculation, in which case, terms of the fourth and fifth power of θ can be added to eq 32 to specifically correct the E_{min} so that $E_{\text{min}} = E_{\text{thresh}}$. This may lead to slightly more accurate results.

4. APPLICATION TO COMPLEX SYSTEMS

4.1. Internal Rotations. Internal rotational (or hindered rotational, HR) degrees of freedom must be treated carefully in any normal-mode analysis. They behave as harmonic oscillators at low temperatures but as free rotors in the high-temperature limit—yet there is no simple analytical expression for their exact partition function. Moreover, internal rotations can lead to anomalous coupling terms in an FD-VPT2 treatment, a result of their motion being poorly described by Cartesian (as opposed to internal, curvilinear) coordinates. In

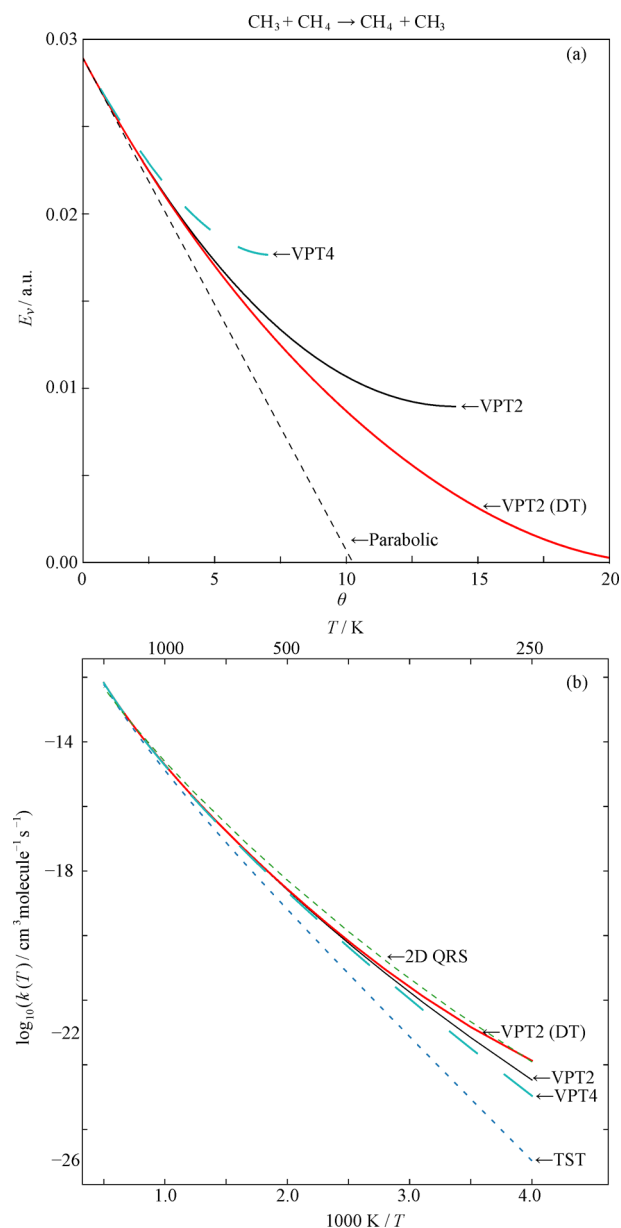


Figure 6. Application of 1D SCTST- θ to the $\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{CH}_3$ reaction. (a) Plot of the VPT2, VPT4, and DT-corrected E_v as a function of θ and (b) plot of calculated reaction rate constants compared to a 2D QRS result.²⁸

addition, the VPT2 expression, while accurate in treating bond stretches and bends, is inaccurate for HR DOF, especially at high temperatures.¹⁵ The HR potential cannot be simply described by a quartic PES function. Careful treatment of HR is of particular importance for reactions in which an internal rotation is present in only the reactant state or the TS; in this case, there can be no cancellation of errors, when the ratio of the partition function is taken to calculate the rate constant. Several approximate treatments for internal rotational DOFs have been suggested.^{48–50} Here, we focus on simple one-dimensional treatments.

An initial assumption is that the HR is separable from the other degrees of freedom. This can be verified by observing a small change in normal-mode frequencies, when the internal rotor is projected out of the Hessian matrix, and it is often found to be valid, owing to the significantly lower frequency

internal rotors typically have.⁴⁰ This separation allows for the HR to be treated with a simple 1D partition function treatment.

Truhlar and co-workers⁵¹ have suggested treating a 1D internal rotation by fitting the rotational energy potential to a cosine Fourier series, from which the partition function is given by a sum over the eigenvalues of the system. This “eigenvalue summation” method requires the 1D torsional PES.

Truhlar et al.⁵¹ have also suggested an approximate 1D partition function (the “tanh” approximation) that moves smoothly from the harmonic (low-temperature) limit, to the free rotor (high-temperature) limit. The “tanh” expression is simply an interpolation function that is accurate in low and high temperatures and smoothly interpolates between them in the intermediate temperatures. Since the computation of the low-temperature harmonic oscillator and high-temperature free rotor partition functions only requires the information on the Hessian matrix, no additional *ab initio* calculations are required.⁴⁰ Note that the expression at intermediate temperatures is only an approximation of the behavior of the function and is not rigorously derived.⁵¹ This is illustrated in Figure 7

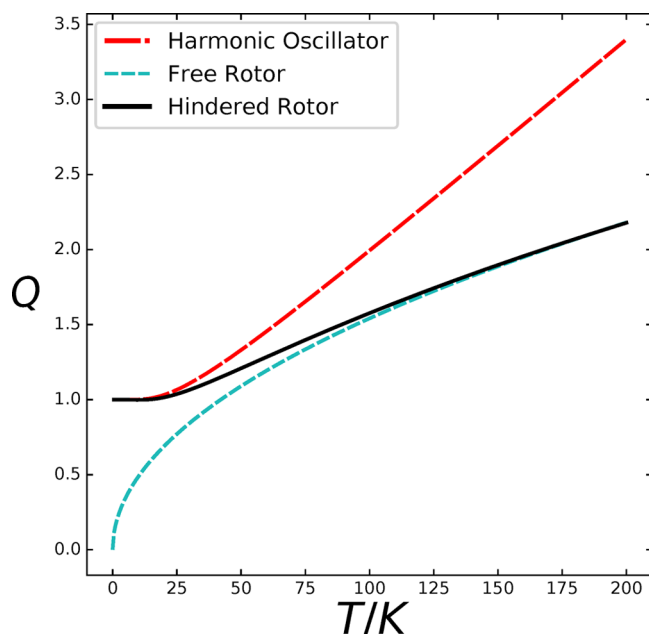


Figure 7. The “tanh” approximation moves smoothly from the harmonic oscillator partition function at low temperature to the free rotor partition function at high temperature. Here, parameters used were from the $\text{H}_3\text{C}-\text{H}-\text{CH}_3$ transition state rotor. Clearly, the harmonic approximation is very poor above 100 K.

for the rotor in the $\text{H}_3\text{C}-\text{H}-\text{CH}_3$ transition state. The effects of this treatment on a reactive system were tested by Burd et al.,⁴⁰ who showed that this provides a qualitative difference to the simple harmonic oscillator approximation and gives excellent agreement with the eigenvalue summation method, as shown in Figure 8a.

Accurate HR treatments are important even for reactions in which both reactant and transition state have internal rotational DOFs. The reaction of H with methanol in the gas phase has been the subject of extensive experimental and theoretical studies.^{39,52–67} A recent study³⁹ showed the harmonic approximation leads to significant errors and worse agreement with experiment. This effect is shown in Figure 8b.

While the 1D solution to HR treatments is straightforward, their treatments in an FD-SCTST calculation are less simple, as they cannot be treated as separable from other degrees of freedom. A treatment by rectilinear (Cartesian) normal modes can lead to unphysically large couplings between it and other modes,^{68,69} but a more sophisticated method, utilizing curvilinear coordinates, is not currently routine. It is likely that more error is introduced by including a hindered rotor in a VPT2 calculation, then by projecting it out of the Hessian matrix and treating it in one dimension.¹⁵ This leaves an $(F - 1)$ dimensional SCTST calculation.

4.2. Application to Degradation of Nerve Agents.

Experimental studies on chemical warfare agents (CWAs) can be difficult to accomplish due to the toxicity of the compounds and also must follow regulation under the Chemical Weapon Convention (CWC). Computational works on CWAs of course do not have the safety issues but face a different challenge, namely, the size of the molecule.^{41,70} For instance, in our recent study of reactions of the nerve agent VX with simulants, *O,S*-diethyl methylphosphonothiolate with 22 atoms (which is approximately half of the size of VX consisting of 42 atoms) has seven internal rotation sites, four of which can produce conformational isomers. In this kind of situation, where multiple HRs are found in the system, approximate treatments to the HR partition function are available.^{48,51,71,72} The most widely used method^{48,71} is based on extending the “tanh” approximation shown in Figures 7 and 8 to a Boltzmann weighted multistructural calculation. We recently showed how to implement these methods in SCTST calculations,^{41,70} in particular, to 1D SCTST.

The 1D SCTST has a natural advantage in adapting these types of improvements on partition function calculations, since the tunneling correction in 1D SCTST is based on the potential energy profile along the reaction mode vibration, which is easily separated from the HR modes. Note that, in these relatively large systems, the HR modes are often strongly coupled with low-frequency bending modes. The approximation made in treating the HR modes by separation can therefore lead to large errors. Since in most reactions, however, the reaction mode corresponds to stretches of chemical bonds, the separation between reaction and HR modes is a good approximation. If the reaction mode is strongly coupled with any HR, then 1D SCTST would not perform well. In such systems, if the coupling between the reaction mode and the HR can be calculated, it is then possible to extend 1D SCTST to RD SCTST with the total reaction rate constant calculated as a sum of HR-state-dependent 1D rate constant. To fully treat HR coupling to low-frequency bends, curvilinear methods may need to be used.

Recently, in a combined theoretical and experimental study⁴¹ of the thermodecomposition of the nerve agent sarin, 1D SCTST with multidimensional HR treatment was successfully applied. The calculated and measured survival percentage of sarin after 0.5 s of reaction time is presented in Figure 9. The 1D SCTST shows excellent agreement to the experimental results. In addition, comparing to the TST results, we can see that even at relatively high temperature (between 600 and 775 K as shown in the figure), quantum tunneling plays an important role in reaction dynamics.

4.3. Application to Atmospheric Reactions. Recent work in our group⁴² studied the application of SCTST to the unimolecular decay of Criegee Intermediates (CIs). CIs play a significant role in atmospheric chemistry,^{73,74} and their decay is

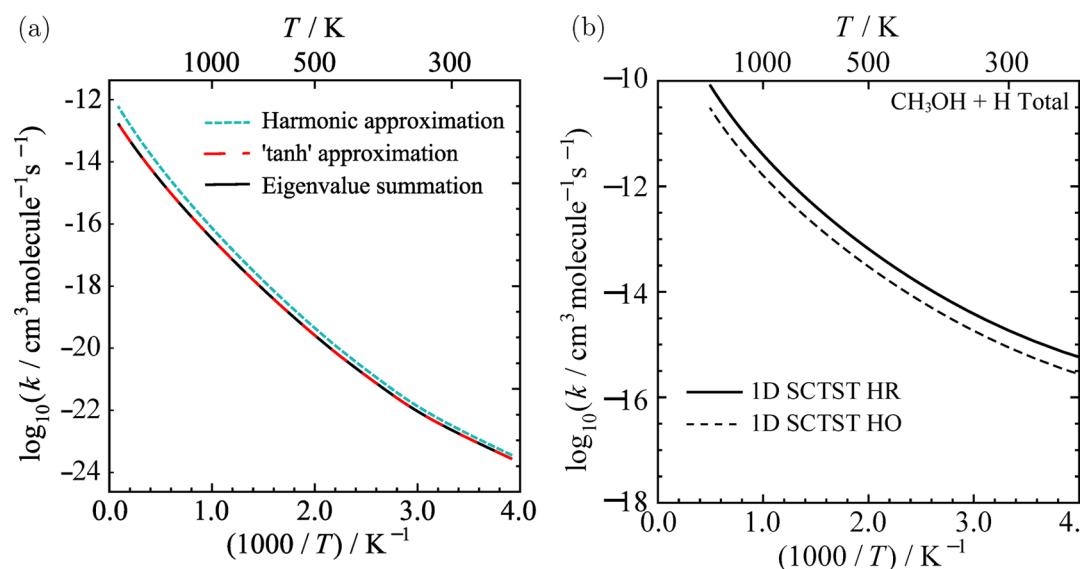


Figure 8. Rate constant comparisons using three treatments of the hindered rotor.

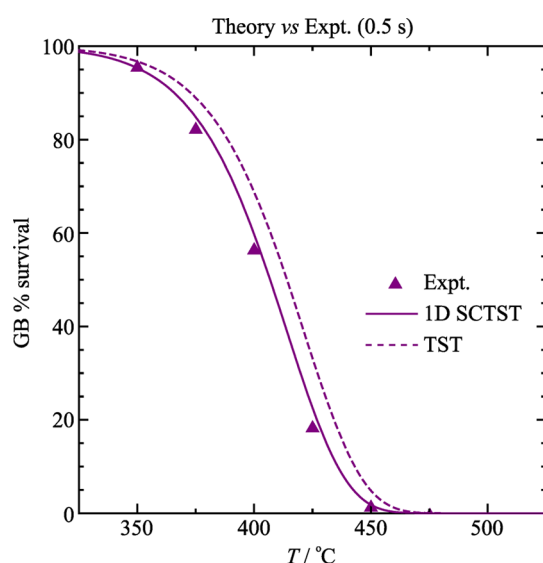


Figure 9. Plot of sarin percentage survival after 0.5 s reaction time vs temperature. Reprinted with permission from ref 41.

thought to be a significant contributor to the OH budget in the atmosphere.⁷⁵ As a result, the reactivity of these species has been the subject of extensive experimental and theoretical studies in recent years.^{76–82}

CIs are typically formed in the atmosphere by the ozonolysis of alkene species. They can then react in a number of different ways, either bimolecularly with “Criegee Scavengers”^{80,83–87} or

with other CI molecules,^{88,89} or they can react unimolecularly.^{90,91} One possible unimolecular decay, shown in Figure 10a, is a key step in the generation of OH radicals, and so it is of particular interest. This reaction can be catalyzed by a single atmospheric water molecule, Figure 10b, although the significance of this pathway under atmospheric conditions is not well-understood. The application of SCTST to this system provides insights not only into the atmospheric chemistry of CIs but also to the applicability of SCTST in reduced dimensions to the reactions of complex systems.

Thermal rate constants for the two processes shown in Figure 10 were calculated using SCTST in full- and one-dimension, as described in detail by ref 42. Single-point energy calculations were performed at the CCSD(T)//MP2/cc-pVTZ level of theory. For optimizing transition states, a *tight* convergence criteria was used.

The results showed good agreement between 1D, FD, and experimental rate constants for the uncatalyzed process, as shown in Figure 11. At 298 K, the 1D and FD rates differed by just 38%. The addition of a water molecule to the transition state reduced the barrier height from 68 to 33 kJ mol^{−1}. In contrast to the uncatalyzed reaction, the catalyzed rate had significantly worse agreement between 1D and FD rates, differing by a factor of 12 at 200 K. In this case, clearly couplings between modes cannot be neglected.

With the FD expression for x_{FF} , given in Section 2 in eq 18, in place of its one-dimensional approximation, can be used to incorporate some anharmonic effects in a cost-effective way. The additional derivatives required can be obtained from the

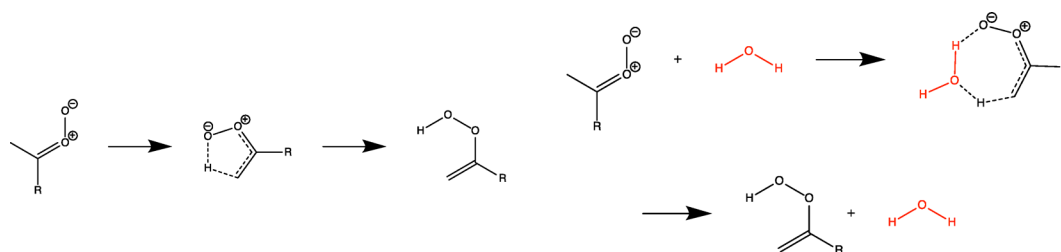


Figure 10. Unimolecular decay pathways of the CI.

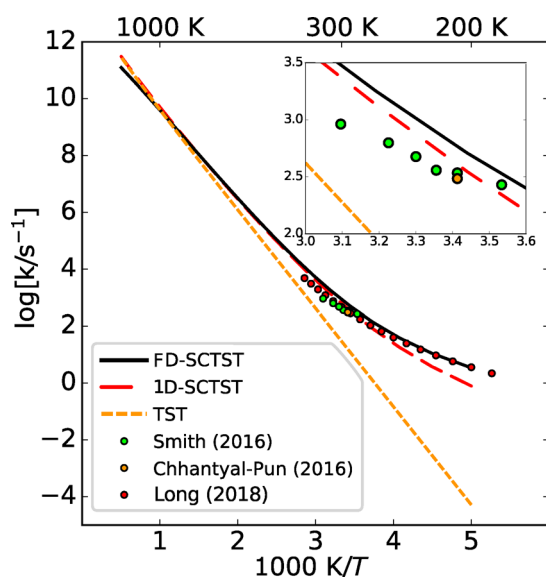


Figure 11. Results for the methylated, uncatalyzed decomposition reaction. Comparison of experimental measurements (Chhantyal-Pun⁹² and Smith⁹³) and other theoretical results (Long)⁹⁴ with TST and SCTST in full and one dimensions (this work). (inset) Clearer comparison of this work to experiment. Reprinted with permission from ref 42.

calculation of two Hessian matrices, displaced along the reaction coordinate. This partially accounts for mode couplings, while maintaining the harmonic spectator mode treatment. The results, shown in Figure 12, show this clearly

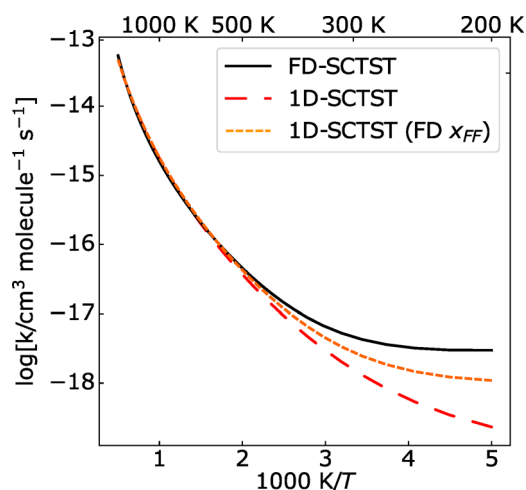


Figure 12. Comparison of FD-SCTST, 1D-SCTST, and 1D-SCTST using the FD value of x_{FF} for the unmethylated, catalyzed reaction. Reprinted with permission from ref 42.

recovers a significant amount of the tunneling, but discrepancies still remain. In the next section we discuss performing SCTST in higher dimensions, $1 < d < F$ to obtain FD quality results, with significantly less computational effort.

4.4. RD SCTST. The contribution of each mode to the tunneling process in the FD calculation can be investigated by studying the values of x_{iF} , as shown for the two Criegee transition states, in Figure 13. These are calculated only in the FD treatment. Clearly, many of the spectator modes do not couple strongly to the reaction mode, and thus their inclusion

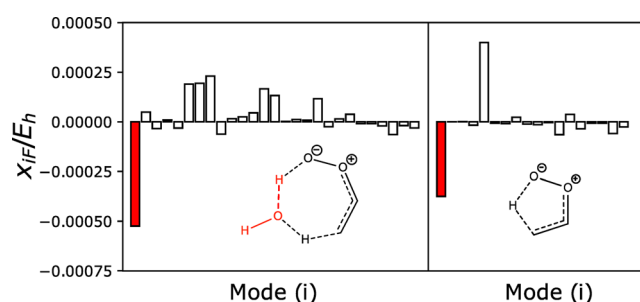


Figure 13. Comparison of values of x_{iF} for the catalyzed and uncatalyzed reaction. The reaction mode anharmonic constant (x_{FF}) is highlighted in red.

in the FD model has little effect on the result. In particular, for the uncatalyzed reaction, only one bound mode is seen to interact strongly with the reaction mode. This suggests that, while the 1D model may be inadequate, the FD model is inefficient, as much computational time is spent calculating coupling constants that are close to zero. RD-SCTST seeks to bridge this gap by treating some, but not all, of the bound modes anharmonically and coupled to the reaction mode. In this way, key couplings can be included, but no time is wasted including modes that are unimportant. RD-SCTST calculations are potentially much more efficient than FD-SCTST, as long as the significant modes can be identified in advance, cheaply. Here we will use the term RD to refer to reduced dimensional methods other than 1D-SCTST.

In principle, RD-SCTST can use any set of degrees of freedom as a basis to form the active space; previous studies have used a bond-breaking and bond-forming bond length basis to perform a two-dimensional calculation for simple hydrogen abstraction reactions.^{17,24} In this work we use the transition-state normal modes as the basis for the active degrees of freedom for simplicity; we will see this allows for straightforward identification of strongly coupled degrees of freedom and can reduce the number of ab initio calculations required.

For a d -dimensional calculation, normal modes of the transition state can be divided into d active modes and $F - d$ spectator modes:

$$\nu = \left\{ \begin{array}{c} \nu_1 \\ \vdots \\ \nu_{F-d} \\ \nu_{F-d+1} \\ \vdots \\ \nu_F \end{array} \right\} \left\{ \begin{array}{l} \text{Spectator Modes} \\ \text{Active Modes} \end{array} \right.$$

The active modes are comprised of the reaction mode and the $d - 1$ “transition” modes. Spectator modes are treated harmonically, as in 1D-SCTST, and so only contribute to the rate through their partition function.

RD-SCTST presents two unique challenges—determining which modes are significant without calculation of the full anharmonic coupling matrix, $\mathbf{X}$, and ensuring a consistent

anharmonic treatment of reactants and products. These are discussed in turn in the subsequent sections.

4.4.1. Selecting Active Modes. Unlike 1D or FD calculations, to perform an RD-SCTST calculation, special care must be taken to decide which, and how many, degrees of freedom to treat as active modes. A poor choice of active modes can lead to wasted computational effort or introduction of errors that are fortuitously canceled in a one-dimensional calculation. Adding in a single transition mode may also overcompensate in one direction and give a worse result than the one-dimensional calculation. This is illustrated in Figure 14.

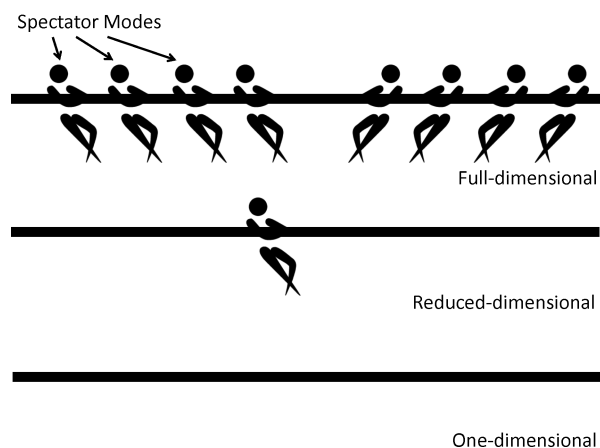


Figure 14. How reduced dimensionality calculations can go wrong, by comparison to modeling a tug-of-war. If the tug of war is close, the 1D model (with no players) is more accurate than the RD model (with one player pulling in one direction). In the same way, including some modes but not others in your calculation can give you the wrong answer. Here, the one-dimensional result is accurate due to a cancellation of errors in the full-dimensional calculation—this is lost in the two-dimensional calculation.

Thus, we seek a cheap, automated method to estimate the coupling of each mode to the reaction mode using, for example, data available from a one-dimensional calculation. A direct way to determine the significant couplings is to study the semidiagonal fourth-order derivatives f_{FFii} and third-order derivatives of the form f_{Fij} which are available from just two Hessians, at geometries displaced along the reaction coordinate, using a method similar to that of Barone:³²

$$f_{FFii} = \frac{H_{ii}(\delta Q_F) + H_{ii}(-\delta Q_F) - 2H_{ii}(0)}{\delta Q_F^2} \quad (35)$$

and

$$f_{Fij} = \frac{H_{ij}(\delta Q_F) - H_{ij}(-\delta Q_F)}{2\delta Q_F} \quad (36)$$

where $H_{ij}(\delta Q)$ is element (i, j) of the Hessian matrix calculated at a geometry displaced from the transition state by a distance δQ . Note this calculation is significantly cheaper than calculating the $2F$ Hessians required for an FD calculation.

With these and the harmonic frequencies, x_{iF} can be approximated by x_{iF}^{approx} :

$$x_{iF}^{\text{approx}} = \frac{1}{4\omega_i\omega_F} \times \left(f_{iFF} + \sum_{j=1}^F \frac{2f_{ijF}^2(\omega_i^2 + \omega_F^2 - \omega_j^2)}{[(\omega_i + \omega_F)^2 - \omega_j^2][(\omega_i - \omega_F)^2 - \omega_j^2]} \right) \quad (37)$$

which serves as an approximation to the exact expression:

$$x_{iF} = \frac{1}{4\omega_i\omega_F} \times \left(f_{iFF} - \sum_{j=1}^F \frac{f_{ij}f_{jFF}}{\omega_j^2} + \sum_{j=1}^F \frac{2f_{ijF}^2(\omega_i^2 + \omega_F^2 - \omega_j^2)}{[(\omega_i + \omega_F)^2 - \omega_j^2][(\omega_i - \omega_F)^2 - \omega_j^2]} \right) \quad (38)$$

where the second term is not present in x_{iF}^{approx} .

In Figure 15, we show that an analysis of the transition states for the catalyzed and uncatalyzed Criegee decays, x_{iF}^{approx}

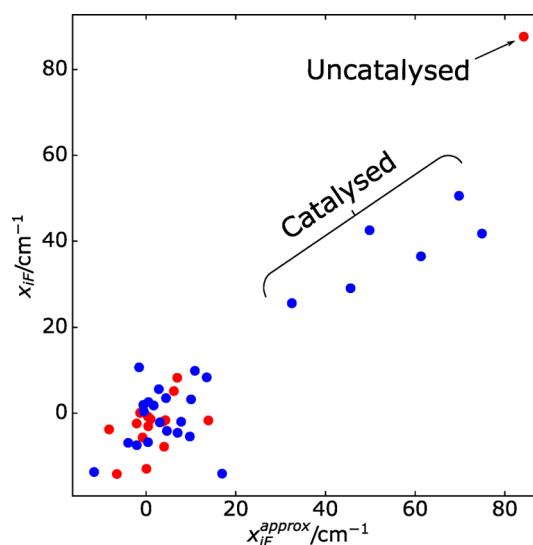


Figure 15. With derivatives available from just two Hessian calculations, x_{iF} can be approximated using eq 37. This allows the significant modes to be identified. Here, modes of the uncatalyzed TS are in red, and modes of the catalyzed TS are in blue.

provides an excellent approximation to x_{iF} and clearly allows for the most significant modes to be distinguished.

Once the key modes have been identified, displaced Hessians along these modes are calculated, allowing for the full calculation of x_{ii} and x_{iF} terms, using derivatives calculated in the same way as in eqs 35 and 36. The anharmonic matrix can thus be constructed using eq 38 and:

$$x_{ii} = \frac{\hbar^2}{16\omega_i^2} \left(f_{iiii} - \sum_{j=1}^i \frac{f_{ij}^2}{\omega_j^2} \frac{8\omega_i^2 - 3\omega_j^2}{4\omega_i^2 - \omega_j^2} \right) \quad (39)$$

As illustrated in Figure 15, for the uncatalyzed reaction, this procedure clearly identifies a single mode is significant ($\omega = 1875 \text{ cm}^{-1}$). For the catalyzed reaction several modes are identified as significant ($\omega = 1745, 1633, 1264, 1650, 1203, 886 \text{ cm}^{-1}$).

4.4.2. Consistent Anharmonic Treatment. Once the significant modes have been determined, the way in which

their anharmonicity is included must be considered. As the rate is calculated by taking the ratio of vibrational partition functions of the transition state and reactants, anharmonicity must be treated in an equivalent way in both systems to ensure the associated errors cancel as much as possible, as shown in eq 3 and Figure 1b.

The most rigorous way to perform an RD calculation is to isolate the $(F - d)$ spectator degrees of freedom through a curvilinear projection²⁸ in the TS and their projection onto the reactant state. This leaves a system with d degrees of freedom with which a d -dimensional SCTST calculation can be done in much the same way as an FD-SCTST calculation is done normally. This method does not limit RD calculations to using a normal mode basis but could be implemented using, for example, a “bond-breaking” and “bond-forming” DOF basis for a 2D calculation.

While theoretically rigorous, this method may be less convenient than others. It requires further Hessians to be calculated in the reactant state, to account for anharmonicity of the projected modes. Moreover, this becomes a less simple extension of 1D-SCTST, which is the desired outcome. The normal mode basis is also a convenient way to look at the reaction for when it is not clear a priori what degrees of freedom may be relevant. Choosing an orthogonal basis (unlike bond stretching bases) also allows for systematic inclusion of more modes, where each can build directly on the computation of the previous result.

When only a few modes are to be considered active ($d \ll F$) it is likely the error in treating these modes anharmonically in only the TS will be small. This is especially true in bimolecular reactions if the active modes partially or completely project onto degrees of freedom in the reactant state that correspond to relative translational/rotational degrees of freedom (whose contribution to the vibrational partition function is zero). Such modes are illustrated in Figure 16 for the $\text{H}_3\text{C}-\text{H}-\text{CH}_3$ transition state. In this work we adopted this simplified approach.

4.4.3. Results. SCTST calculations were performed in RD for the two Criegee decay reactions using the modes determined as most significant by the method described in Section 4.3.1. The results are shown in Figure 17.

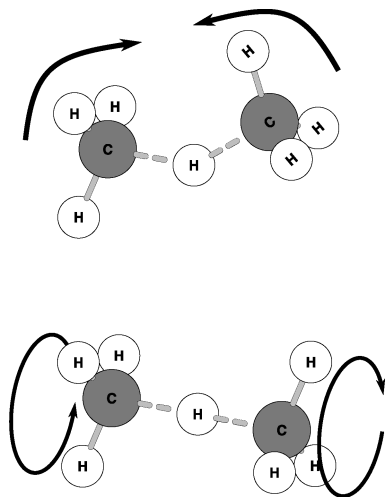


Figure 16. Some modes that correspond to relative translational and rotational modes in the reactant state.

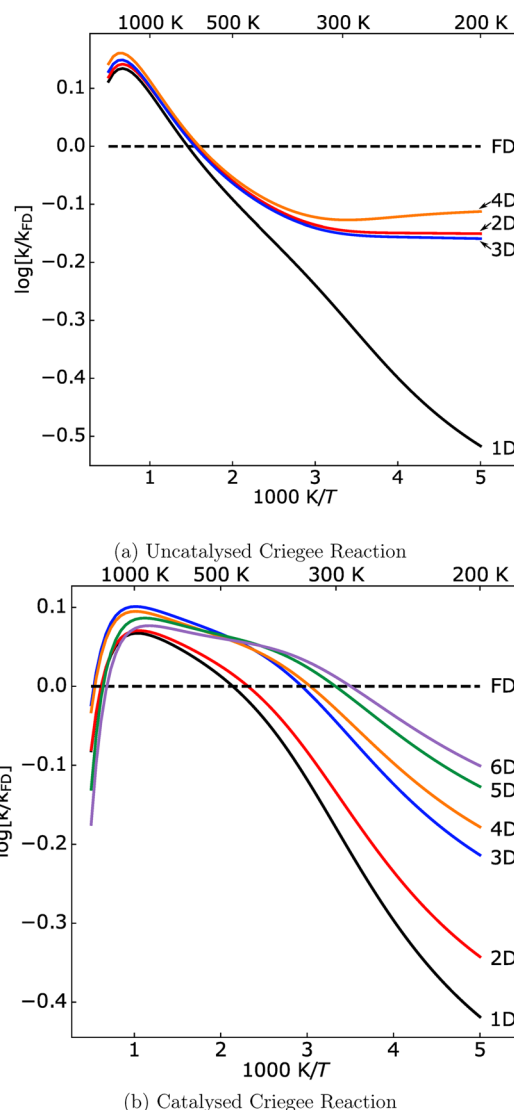


Figure 17. Effect of including additional degrees of freedom in the tunneling calculations. The modes selected are those with the largest values of x_{IF} .

From Figure 17a it is clear that adding in just a single bound mode gives a significant improvement to the rate. Whereas 1D SCTST is a factor of 3 smaller than the FD rate at 200 K, the 2D calculation differs by just 40%. As expected, adding in the next most significant modes to the active space has only a small additional effect on the rate. For the catalyzed reaction, multiple modes contribute significantly to the tunneling process, and so a number of modes are required to achieve a similar accuracy to that of the uncatalyzed reaction. The six-dimensional reaction lies within 30% of the FD rate at all temperatures evaluated.

The development of RD-SCTST provides an extremely powerful tool for calculating SCTST rate constants in an efficient way, provided only a small number of bound modes interact. Further work is required to determine how common systems like this are.

4.5. Understanding Corner-Cutting Included in SCTST. Semiclassical rate theories are able to simplify the calculation of rate constants for multidimensional reactions by defining distinct tunneling pathways through which the reactive flux is calculated. The height and width of the

pathway barrier very sensitively determine the tunneling probability. The way in which these paths are determined from a multidimensional PES varies between methods. A simple choice of pathway is the minimum energy path (MEP) from the reactant to product configurations, as illustrated in Figure 18a, which is easily determined from a steepest descent

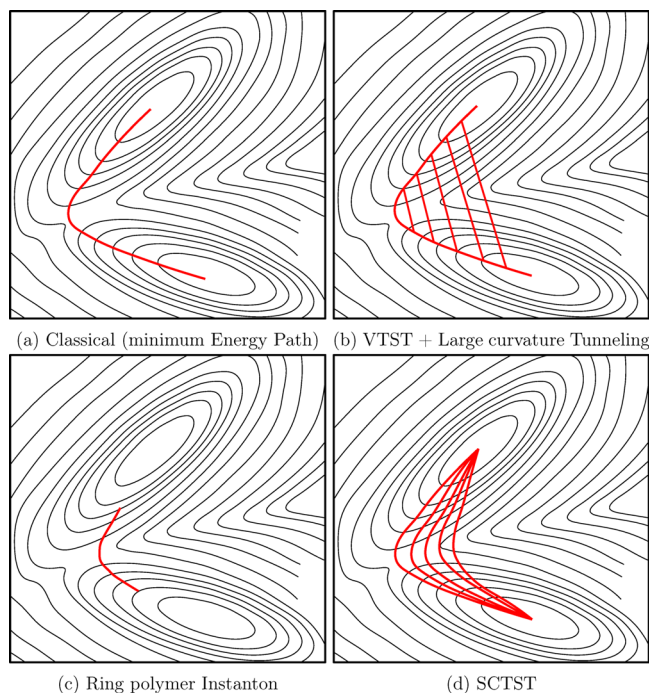


Figure 18. Illustration of how corner-cutting effects are accounted for in three semiclassical methods

calculation from the TS. It is well-known that tunneling through the MEP often underestimates reaction probabilities due to neglecting corner-cutting effects. For reactions whose MEP exhibits large curvature, the reaction may dominantly proceed through a path with a higher but narrower barrier, “cutting the corner” of the MEP. This effect, known as “corner-cutting”, is most significant for “heavy-light-heavy” type of reactions, in which a light atom is transferred between two heavy centers.

The effects of corner-cutting on the dynamics of chemical reactions has been studied extensively,^{95–103} where a key challenge is determining the location on the PES of the important pathways.

Truhlar and co-workers have developed methods to incorporate multidimensional tunneling effects in variational Transition State Theory (VTST), accounting for corner-cutting effects. In the small-curvature tunneling approximation (SCT), corner-cutting is accounted for by reducing the effective mass of the reaction coordinate, equivalent to shortening the tunneling path.^{3,104,105} The new effective mass is determined from the reaction-path curvature. The large-curvature tunneling (LCT) approximation^{98,99,102,106,107} includes tunneling paths that are straight lines that connect the reactant and product wells, as illustrated in Figure 18b. These paths can also connect configurations in different vibrational states. In microcanonically optimized tunneling VTST (μ OMT), the tunneling is calculated as the maximum of the SCT and LCT probabilities. Finally, the least-action tunneling

model (LAT)^{106,108} considers a large number of paths between the limits of the MEP and LCT paths.

Ring-polymer instanton rate theory^{9,109} determines a single dominant tunneling pathway by finding the path that represents a saddle point in the ring-polymer action, shown in Figure 18c. The instanton can therefore be expected to accurately capture the corner-cutting effects of the PES. While this typically requires knowledge of a substantial part of the PES, recent advances, using machine learning methods, have reduced the number of ab initio calculations substantially.¹¹⁰

In this section we show that SCTST can be interpreted as a sum of 1D-SCTST rate constants, each corresponding to a different effective barrier height and width. These barriers can be associated with distinct corner-cutting trajectories, as illustrated in Figure 18d. This interpretation gives a physical picture of how corner-cutting paths are incorporated into SCTST. Note that, since the current derivation of SCTST does not include any variational effects, here we only focus on the computation of quantum tunneling correction for different methods. We therefore assume that the systems shown in any of the illustrated figures in this section do not have significant variational effects; that is, the TS is the maximum on the ZPE corrected potential.

The connection between FD-SCTST and individual 1D-SCTST paths is clear from comparing the key formulas for the rate. In 1D-SCTST, the action variable of the F th degree of freedom can be written as a function of the energy E_v , the barrier height ΔV_b and the barrier frequency ω_F :

$$\theta_{1D}(E_v, \Delta V_f, \omega_F) = \pi \frac{i\omega_F + [-\omega_F^2 + 4x_{FF}(\Delta V_f + G_0 - E_v)]^{1/2}}{2x_{FF}} \quad (40)$$

In the case where the barrier height is replaced by an effective barrier

$$V_{\{n\}'}^{\text{eff}} = \Delta V_f + E_{\{n\}'}, \quad (41)$$

with

$$E_{\{n\}'} = \sum_{i=1}^F \hbar\omega_i(n_i + 0.5) + \sum_{i < j}^F x_{ij}(n_i + 0.5)(n_j + 0.5) + G_0 \quad (42)$$

and the frequency replaced with an effective frequency

$$\omega_{\{n\}'}^{\text{eff}} = \omega_F + \sum_{i=1}^{F-1} x_{iF}(n_i + 0.5) \quad (43)$$

the FD expression for θ is returned:

$$\begin{aligned} \theta_{1D}(E_v, V_{\{n\}'}^{\text{eff}}, \omega_{\{n\}'}^{\text{eff}}) &= \pi \\ &\times \frac{-\Omega_{\{n\}'}^2 + [\Omega_{\{n\}'}^2 + 4x_{FF}(\Delta V_f + G_0 - E_{\{n\}'} - E_v)]^{1/2}}{2x_{FF}} \\ &= \theta_{FD}(E_v, \{n\}') \end{aligned} \quad (44)$$

The state-dependent reaction probability is then given by

$$P_{\{n\}'}(E_v) = (1 + \exp[2\theta_{1D}(E_v, V_{\{n\}'}^{\text{eff}}, \omega_{\{n\}'}^{\text{eff}})])^{-1} \quad (45)$$

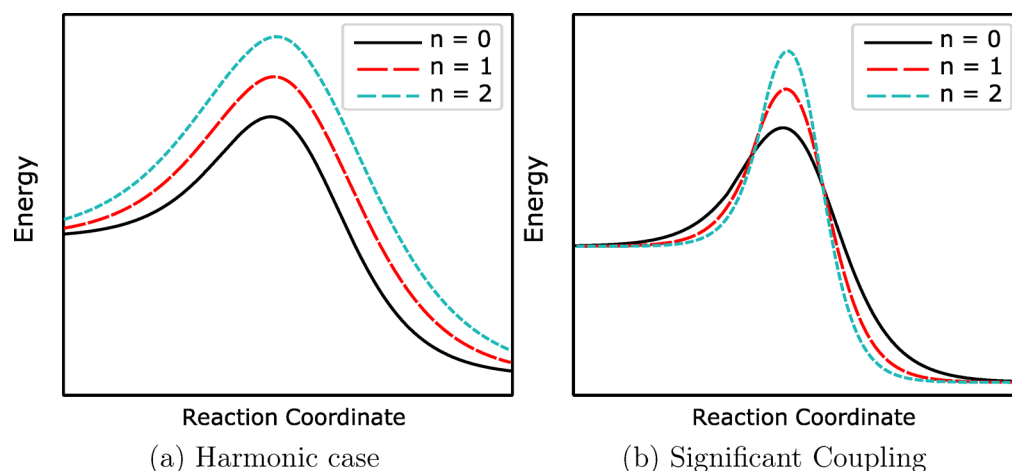


Figure 19. Illustrative paths for different active mode quantum numbers.

and the cumulative reaction probability is given by the sum over all state-dependent reaction probabilities at the energy E_v :

$$N(E_v) = \sum_{\{n\}'} P_{\{n\}'}(E_v) \quad (46)$$

The FD rate can thus be interpreted as a sum over one-dimensional barriers, with an effective barrier height and frequency given by eqs 41 and 43:

$$k_{\text{FD}} = \sum_{\{n\}'} k_{\text{1D}}(V_{\{n\}'}^{\text{eff}}, \omega_{\{n\}'}^{\text{eff}}) \quad (47)$$

where eq 47 is equally applicable to thermal and microcanonical rate constants. Each effective barrier corresponds to a vibrational configuration of the transition state. As a mode's quantum number n_i increases, the effective barrier increases as described by eq 41. If $x_{iF} > 0$, the corresponding effective frequency will also increase, representing a narrower barrier. These distinct barriers can then be interpreted as corner-cutting paths. This is illustrated in Figure 19, where barriers for a harmonic system are compared to a strongly anharmonic system. In the harmonic case, $x_{iF} = 0$, multiple paths are still summed over, and each subsequent barrier is higher than the last, but no barrier narrowing is seen.

It is interesting to ask whether these distinct pathways can be located on a given PES, to give insight about how the reaction proceeds. In general, this is only possible on a surface for which VPT2 is exact. In 1D, this corresponds to a harmonic oscillator, or a Morse potential. In multiple dimensions, however, the only simple potential for which VPT2 is exact is an uncoupled harmonic oscillator function. For anharmonic potentials, these paths do not exist on the "real" PES, but rather on a "pseudosurface" for which VPT2 is exact, and whose third and semidiagonal fourth-order derivatives match those of the real surface.

5. CONCLUSION

Semiclassical Transition State Theory was first formulated in the 1970s by W. H. Miller, and the first realistic applications of the theory were made by Miller and co-workers in the 1990s. However, it is only recently that the accuracy and power of this method for calculating tunneling effects on chemical reaction rate constants has been fully realized.

Comparisons with quantum reactive scattering calculations have demonstrated the accuracy of the method, even for

reactions such as H atom transfer between two heavy molecules that have caused difficulties in the past. Furthermore, the advances in quantum chemistry methods have allowed for the efficient and accurate calculation of the required quantities that are needed to apply the method—the second, third, and fourth derivatives of the potential surface at the transition state of the reaction. In addition, the recent work has shown that only a small number of accurate ab initio computations are needed beyond those required in applications of regular transition state theory. The method also has the distinct advantage that a global PES does not need to be constructed.

This Feature Article has discussed some of the recent developments and applications of the SCTST performed in our own research group. In particular, we show that a reduced dimensional form of the SCTST is particularly efficient and reduces the required electronic structure calculations to an absolute minimum. We also describe how an alternative formulation of SCTST, called SCTST- θ , has advantages in allowing for straightforward applications of the SCTST for any form of the potential expansion at the transition state. When combined with a deep tunneling correction, which imposes the correct exothermicity or endothermicity on a reaction, this approach becomes particularly powerful.

We also describe some applications of the SCTST to more complex systems. Polyatomic molecules often have many internal rotations and torsions, and we examine computational procedures for dealing with these types of modes. We also describe some applications of the method to tunneling in unimolecular reactions. A particularly timely application of relevance to global security is to the degradation of chemical nerve agents, molecules that are very difficult to study in the laboratory. Another application described is the unimolecular decay of the atmospherically important Criegee intermediates, which also occur through a hydrogen atom transfer via a cyclic transition state. Here it is also shown that a single water molecule significantly catalyzes the reaction.

Using the Criegee intermediates as an example we also describe computational procedures for examining when the reduced dimensionality approach is most applicable and how to select the most active vibrational modes taking part in a chemical reaction. We also explain in some detail why the SCTST works so well for H atom exchange reactions of the

heavy-light-heavy form, where corner-cutting of the potential surface comes into play.

It has become clear that the SCTST is a very reliable and computationally inexpensive approach for treating quantum mechanical tunneling in chemical reactions. The method is a direct way of turning a minimal number of electronic structure calculations into rate constants. It is an important addition to the repertoire of *ab initio* methods that are enabling theory to make accurate and useful predictions on chemical reactions.

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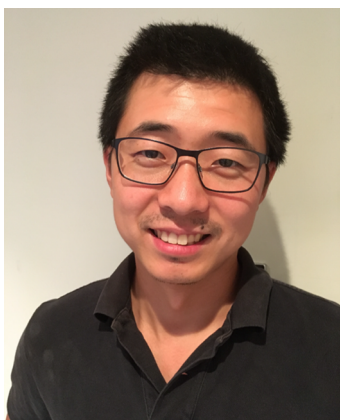
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Notes

The authors declare no competing financial interest.

Biographies



Xiao Shan undertook his Ph.D. (2006–2010) at the University of Manchester with Professor J. N. L. Connor. He has worked as a postdoctoral research associate in Professor D. C. Clary's group in the Chemistry Department at the University of Oxford since 2011. His main research interest is in application of quantum and semiclassical methods to chemical reactions involving typically more than 10 atoms and in numerical analyses to differential cross sections of reactive scattering processes.



Tim Burd joined the Department of Chemistry at the University of Oxford in 2016, and he is currently pursuing a D.Phil. degree in theoretical chemistry under the supervision of Prof. Sir David Clary. His research focuses on the development of efficient rate theories for

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David Clary has been President of Magdalen College Oxford since 2005. He also directs a research group in the Chemistry Department at Oxford University working on the quantum theory of chemical reactions. He was the first Chief Scientific Advisor to the UK Foreign and Commonwealth Office (2009–2013) and was President of the Faraday Division of the Royal Society of Chemistry (2006–2009). He was elected as a Fellow of the Royal Society in 1997 and was knighted by the Queen for services to international science in 2016.

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REFERENCES

- (1) Suleimanov, Y. V.; Aoiz, F. J.; Guo, H. Chemical Reaction Rate Coefficients from Ring Polymer Molecular Dynamics: Theory and Practical Applications. *J. Phys. Chem. A* **2016**, *120*, 8488–8502.
- (2) *Faraday Discussions: Reaction Rate Theory*; Royal Society of Chemistry, 2016; Vol. 195.
- (3) Bao, J. L.; Truhlar, D. G. Variational Transition State Theory: Theoretical Framework and Recent Developments. *Chem. Soc. Rev.* **2017**, *46*, 7548–7596.
- (4) Clary, D. C. Spiers Memorial Lecture Introductory Lecture: Quantum Dynamics of Chemical Reactions. *Faraday Discuss.* **2018**, *212*, 9–32.
- (5) Fu, B.; Shan, X.; Zhang, D. H.; Clary, D. C. Recent Advances in Quantum Scattering Calculations on Polyatomic Bimolecular Reactions. *Chem. Soc. Rev.* **2017**, *46*, 7625–7649.
- (6) Zhao, B.; Guo, H. State-to-State Quantum Reactive Scattering in Four-Atom Systems. *WIREs Comput. Mol. Sci.* **2017**, *7*, No. e1301.
- (7) Truhlar, D. G.; Garrett, B. C.; Klippenstein, S. J. Current Status of Transition-State Theory. *J. Phys. Chem.* **1996**, *100*, 12771–12800.
- (8) Habershon, S.; Manolopoulos, D. E.; Markland, T. E.; Miller, T. F., III Ring-Polymer Molecular Dynamics: Quantum Effects in Chemical Dynamics from Classical Trajectories in an Extended Phase Space. *Annu. Rev. Phys. Chem.* **2013**, *64*, 387–413.
- (9) Richardson, J. O. Ring-Polymer Instanton Theory. *Int. Rev. Phys. Chem.* **2018**, *37*, 171–216.

- (10) Schlegel, H. B. Exploring Potential Energy Surfaces for Chemical Reactions: An Overview of Some Practical Methods. *J. Comput. Chem.* **2003**, *24*, 1514–1527.
- (11) Qu, C.; Yu, Q.; Bowman, J. M. Permutationally Invariant Potential Energy Surfaces. *Annu. Rev. Phys. Chem.* **2018**, *69*, 151–175.
- (12) Miller, W. H. Semiclassical Limit of Quantum Mechanical Transition State Theory for Nonseparable Systems. *J. Chem. Phys.* **1975**, *62*, 1899–1906.
- (13) Miller, W. H. Semi-Classical Theory for Non-Separable Systems: Construction of “Good” Action-Angle Variables for Reaction Rate Constants. *Faraday Discuss. Chem. Soc.* **1977**, *62*, 40–46.
- (14) Nguyen, T. L.; Stanton, J. F.; Barker, J. R. *Ab Initio* Reaction Rate Constants Computed Using Semiclassical Transition-State Theory: $\text{HO} + \text{H}_2 \rightarrow \text{H}_2\text{O} + \text{H}$ and Isotopologues. *J. Phys. Chem. A* **2011**, *115*, 5118–5126.
- (15) Nguyen, T. L.; Stanton, J. F. High-Level Theoretical Study of the Reaction Between Hydroxyl and Ammonia: Accurate Rate Constants from 200 to 2500 K. *J. Chem. Phys.* **2017**, *147*, 152704.
- (16) Greene, S. M.; Shan, X.; Clary, D. C. Quantum Scattering and Semiclassical Transition State Theory Calculations on Chemical Reactions of Polyatomic Molecules in Reduced Dimensions. *Adv. Chem. Phys.* **2018**, *163*, 117–149.
- (17) Greene, S. M.; Shan, X.; Clary, D. C. Reduced-Dimensionality Semiclassical Transition State Theory: Application to Hydrogen Atom Abstraction and Exchange Reactions of Hydrocarbons. *J. Phys. Chem. A* **2015**, *119*, 12015–12027.
- (18) Bowman, J. M. Reduced Dimensionality Theory of Quantum Reactive Scattering. *J. Phys. Chem.* **1991**, *95*, 4960–4968.
- (19) Kerkeni, B.; Clary, D. C. *Ab Initio* Rate Constants from Hyperspherical Quantum Scattering: Application to $\text{H} + \text{CH}_4 \rightarrow \text{H}_2 + \text{CH}_3$. *J. Chem. Phys.* **2004**, *120*, 2308–2318.
- (20) Banks, S. T.; Clary, D. C. Chemical Reaction Surface Vibrational Frequencies Evaluated in Curvilinear Internal Coordinates: Application to $\text{H} + \text{CH}_4 \rightleftharpoons \text{H}_2 + \text{CH}_3$. *J. Chem. Phys.* **2009**, *130*, 024106.
- (21) von Horsten, H. F.; Banks, S. T.; Clary, D. C. An Efficient Route to Thermal Rate Constants in Reduced Dimensional Quantum Scattering Simulations: Applications to the Abstraction of Hydrogen from Alkanes. *J. Chem. Phys.* **2011**, *135*, 094311.
- (22) Shan, X.; Clary, D. C. Quantum Effects in the Abstraction Reaction by H Atoms of Primary and Secondary Hydrogens in $n\text{-C}_4\text{H}_{10}$: A Test of a New Potential Energy Surface Construction Method. *Phys. Chem. Chem. Phys.* **2013**, *15*, 1222–1231.
- (23) Shan, X.; Clary, D. C. Quantum Dynamics of the Abstraction Reaction of H with Cyclopropane. *J. Phys. Chem. A* **2014**, *118*, 10134–10143.
- (24) Greene, S. M.; Shan, X.; Clary, D. C. An Investigation of One-Versus Two-Dimensional Semiclassical Transition State Theory for H Atom Abstraction and Exchange Reactions. *J. Chem. Phys.* **2016**, *144*, 084113.
- (25) Miller, W. H. Recent Advances in Quantum Mechanical Reactive Scattering Theory, Including Comparison of Recent Experiments with Rigorous Calculations of State-to-State Cross Sections for the $\text{H}/\text{D} + \text{H}_2 \rightarrow \text{H}_2/\text{HD} + \text{H}$ Reactions. *Annu. Rev. Phys. Chem.* **1990**, *41*, 245–281.
- (26) Miller, W. H.; Hernandez, R.; Handy, N. C.; Jayatilaka, D.; Willetts, A. *Ab Initio* Calculation of Anharmonic Constants for a Transition State, with Application to Semiclassical Transition State Tunneling Probabilities. *Chem. Phys. Lett.* **1990**, *172*, 62–68.
- (27) Barker, J. R.; Nguyen, T. L.; Stanton, J. F. Kinetic Isotope Effects for $\text{Cl} + \text{CH}_4 \rightleftharpoons \text{HCl} + \text{CH}_3$ Calculated Using *Ab Initio* Semiclassical Transition State Theory. *J. Phys. Chem. A* **2012**, *116*, 6408–6419.
- (28) Banks, S. T.; Tautermann, C. S.; Remmert, S. M.; Clary, D. C. An Improved Treatment of Spectator Mode Vibrations in Reduced Dimensional Quantum Dynamics: Application to the Hydrogen Abstraction Reactions $\mu + \text{CH}_4$, $\text{H} + \text{CH}_4$, $\text{D} + \text{CH}_4$, and $\text{CH}_3 + \text{CH}_4$. *J. Chem. Phys.* **2009**, *131*, 044111.
- (29) Meng, Q. Ring-Polymer Molecular Dynamics Study on Rate Coefficients of Hydrogen Abstraction of Methane: A Reduced-Dimensional Model. *Chem. Phys. Lett.* **2018**, *706*, 383–387.
- (30) Beyer, A. N.; Richardson, J. O.; Knowles, P. J.; Rommel, J.; Althorpe, S. C. Quantum Tunneling Rates of Gas-Phase Reactions from On-the-Fly Instanton Calculations. *J. Phys. Chem. Lett.* **2016**, *7*, 4374.
- (31) Hoy, A.; Mills, I.; Strey, G. Anharmonic Force Constant Calculations. *Mol. Phys.* **1972**, *24*, 1265–1290.
- (32) Barone, V. Anharmonic Vibrational Properties by a Fully Automated Second-Order Perturbative Approach. *J. Chem. Phys.* **2005**, *122*, 014108.
- (33) Greene, S. M.; Shan, X.; Clary, D. C. Rate Constants of Chemical Reactions from Semiclassical Transition State Theory in Full and One Dimension. *J. Chem. Phys.* **2016**, *144*, 244116.
- (34) Nguyen, T. L.; Barker, J. R. Sums and Densities of Fully Coupled Anharmonic Vibrational States: A Comparison of Three Practical Methods. *J. Phys. Chem. A* **2010**, *114*, 3718–3730.
- (35) Wang, F.; Landau, D. P. Efficient, Multiple-Range Random Walk Algorithm to Calculate the Density of States. *Phys. Rev. Lett.* **2001**, *86*, 2050–2053.
- (36) Wang, F.; Landau, D. P. Determining the Density of States for Classical Statistical Models: A Random Walk Algorithm to Produce a Flat Histogram. *Phys. Rev. E: Stat. Phys., Plasmas, Fluids, Relat. Interdiscip. Top.* **2001**, *64*, 056101.
- (37) Basire, M.; Parneix, P.; Calvo, F. Quantum Anharmonic Densities of States Using the Wang-Landau Method. *J. Chem. Phys.* **2008**, *129*, 081101.
- (38) Aieta, C.; Gabas, F.; Ceotto, M. An Efficient Computational Approach for the Calculation of the Vibrational Density of States. *J. Phys. Chem. A* **2016**, *120*, 4853–4862.
- (39) Shan, X.; Clary, D. C. Application of One-Dimensional Semiclassical Transition State Theory to the $\text{CH}_3 + \text{H} \rightleftharpoons \text{CHO}_2\text{H}/\text{CHO}_3 + \text{H}_2$ Reactions. *Philos. Trans. R. Soc., A* **2018**, *376*, 20170147.
- (40) Burd, T. A. H.; Shan, X.; Clary, D. C. Tunneling and the Kinetic Isotope Effect in $\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{CH}_3$: An Application of Semiclassical Transition State Theory. *Chem. Phys. Lett.* **2018**, *693*, 88–94.
- (41) Shan, X.; Vincent, J. C.; Kirkpatrick, S.; Walker, M. D.; Sambrook, M. R.; Clary, D. C. A Combined Theoretical and Experimental Study of Sarin (GB) Decomposition at High Temperatures. *J. Phys. Chem. A* **2017**, *121*, 6200–6210.
- (42) Burd, T. A. H.; Shan, X.; Clary, D. C. Catalysis and Tunneling in the Unimolecular Decay of Criegee Intermediates. *Phys. Chem. Chem. Phys.* **2018**, *20*, 25224.
- (43) Wagner, A. F. Improved Multidimensional Semiclassical Tunneling Theory. *J. Phys. Chem. A* **2013**, *117*, 13089–13100.
- (44) Stanton, J. F. Semiclassical Transition-State Theory Based on Fourth-Order Vibrational Perturbation Theory: The Symmetrical Eckart Barrier. *J. Phys. Chem. Lett.* **2016**, *7*, 2708–2713.
- (45) Goel, P.; Stanton, J. F. Semiclassical Transition State Theory Based on Fourth Order Vibrational Perturbation Theory: Model System Studies beyond Symmetric Eckart Barrier. *J. Chem. Phys.* **2018**, *149*, 134109.
- (46) Hernandez, R.; Miller, W. H. Semiclassical Transition State Theory: A New Perspective. *Chem. Phys. Lett.* **1993**, *214*, 129–136.
- (47) Fermann, J. T.; Auerbach, S. Modeling Proton Mobility in Acidic Zeolite Clusters: II. Room Temperature Tunneling Effects from Semiclassical Rate Theory. *J. Chem. Phys.* **2000**, *112*, 6787.
- (48) Ellingson, B. A.; Lynch, V. A.; Mielke, S. L.; Truhlar, D. G. Statistical Thermodynamics of Bond Torsional Modes: Tests of Separable, Almost-Separable, and Improved Pitzer-Gwinn Approximations. *J. Chem. Phys.* **2006**, *125*, 084305.
- (49) Zheng, J.; Truhlar, D. G. Multi-Path Variational Transition State Theory for Chemical Reaction Rates of Complex Polyatomic Species: Ethanol + OH Reactions. *Faraday Discuss.* **2012**, *157*, 59–88.
- (50) Lin, C. Y.; Izgorodina, E. I.; Coote, M. L. How Accurate Are Approximate Methods for Evaluating Partition Functions for Hindered Internal Rotations? *J. Phys. Chem. A* **2008**, *112*, 1956–1964.

- (51) Chuang, Y.-Y.; Truhlar, D. G. Statistical Thermodynamics of Bond Torsional Modes. *J. Chem. Phys.* **2000**, *112*, 1221–1228.
- (52) Blowers, P.; Ford, L.; Masel, R. *Ab Initio* Calculations of the Reactions of Hydrogen with Methanol: A Comparison of the Role of Bond Distortions and Pauli Repulsions on the Intrinsic Barriers for Chemical Reactions. *J. Phys. Chem. A* **1998**, *102*, 9267–9277.
- (53) Kerkeni, B.; Clary, D. C. Kinetic Isotope Effects in the Reactions of D Atoms with CH₄, C₂H₆, and CH₃OH: Quantum Dynamics Calculations. *J. Phys. Chem. A* **2004**, *108*, 8966–8972.
- (54) Cribb, P. H.; Dove, J. E.; Yamazaki, S. A Kinetic Study of the Pyrolysis of Methanol Using Shock Tube and Computer Simulation Techniques. *Combust. Flame* **1992**, *88*, 169–185.
- (55) Lendvay, G.; Bérces, T.; Márta, F. An *ab Initio* Study of the Three-Channel Reaction between Methanol and Hydrogen Atoms: BAC-MP4 and Gaussian-2 Calculations. *J. Phys. Chem. A* **1997**, *101*, 1588–1594.
- (56) Zhu, T.; Li, J.; Liotard, D. A.; Cramer, C. J.; Truhlar, D. G. Analytical Energy Gradients of a Self-Consistent Reaction-Field Solvation Model Based on CM2 Atomic Charges. *J. Chem. Phys.* **1999**, *110*, 5503–5513.
- (57) Pu, J.; Ma, S.; Gao, J.; Truhlar, D. G. Small Temperature Dependence of the Kinetic Isotope Effect for the Hydride Transfer Reaction Catalyzed by *Escherichia coli* Dihydrofolate Reductase. *J. Phys. Chem. B* **2005**, *109*, 8551–8556.
- (58) Tsang, W. Chemical Kinetic Data Base for Combustion Chemistry. Part 2. Methanol. *J. Phys. Chem. Ref. Data* **1987**, *16*, 471–508.
- (59) Chuang, Y.-Y.; Truhlar, D. G. Nonequilibrium Solvation Effects for a Polyatomic Reaction in Solution. *J. Am. Chem. Soc.* **1999**, *121*, 10157–10167.
- (60) Baulch, D. L.; Bowman, C. T.; Cobos, C. J.; Cox, R. A.; Just, T.; Kerr, J. A.; Pilling, M. J.; Stocker, D.; Troe, J.; Tsang, W.; et al. Evaluated Kinetic Data for Combustion Modeling: Supplement II. *J. Phys. Chem. Ref. Data* **2005**, *34*, 757–1397.
- (61) Li, S. C.; Williams, F. A. Experimental and Numerical Studies of Two-Stage Methanol Flames. *Symp. Combust., [Proc.]* **1996**, *26*, 1017–1024.
- (62) Meana-Pañeda, R.; Truhlar, D. G.; Fernández-Ramos, A. High-Level Direct-Dynamics Variational Transition State Theory Calculations Including Multidimensional Tunneling of the Thermal Rate Constants, Branching Ratios, and Kinetic Isotope Effects of the Hydrogen Abstraction Reactions from Methanol by Atomic Hydrogen. *J. Chem. Phys.* **2011**, *134*, 094302.
- (63) Meagher, J. F.; Kim, P.; Lee, J. H.; Timmons, R. B. Kinetic Isotope Effects in the Reactions of Hydrogen and Deuterium Atoms with Dimethyl Ether and Methanol. *J. Phys. Chem.* **1974**, *78*, 2650–2657.
- (64) Aronowitz, D.; Naegeli, D. W.; Glassman, I. Kinetics of the Pyrolysis of Methanol. *J. Phys. Chem.* **1977**, *81*, 2555–2559.
- (65) Hoyeremann, K.; Sievert, R.; Wagner, H. G. Mechanism of the Reaction of H Atoms with Methanol. *Berichte der Bunsengesellschaft für Phys. Chemie* **1981**, *85*, 149–153.
- (66) Carvalho, E. F. V.; Barauna, A. N.; Machado, F. B. C.; Roberto-Neto, O. Theoretical Calculations of Energetics, Structures, and Rate Constants for the H + CH₃OH Hydrogen Abstraction Reactions. *Chem. Phys. Lett.* **2008**, *463*, 33–37.
- (67) Jodkowski, J. T.; Rayez, M.-T.; Rayez, J.-C.; Bérces, T.; Dóbe, S. Theoretical Study of the Kinetics of the Hydrogen Abstraction from Methanol. 3. Reaction of Methanol with Hydrogen Atom, Methyl, and Hydroxyl Radicals. *J. Phys. Chem. A* **1999**, *103*, 3750–3765.
- (68) Miller, W. H. Quantum Dynamics of Complex Molecular Systems. *Proc. Natl. Acad. Sci. U. S. A.* **2005**, *102*, 6660–6664.
- (69) Jung, J. O.; Gerber, R. B. Vibrational Wave Functions and Spectroscopy of (H₂O)_n, n = 2,3,4,5: Vibrational Self-Consistent Field with Correlation Corrections. *J. Chem. Phys.* **1996**, *105*, 10332–10348.
- (70) Shan, X.; Sambrook, M. R.; Clary, D. C. Theoretical Study of Gas-Phase Unimolecular Decomposition of Simulants of the Nerve Agent VX. *J. Phys. Chem. A* **2019**, *123*, 59–72.
- (71) Zheng, J.; Truhlar, D. G. Kinetics of Hydrogen-Transfer Isomerizations of Butoxyl Radicals. *Phys. Chem. Chem. Phys.* **2010**, *12*, 7782–7793.
- (72) Zheng, J.; Yu, T.; Papajak, E.; Alecu, I. M.; Mielke, S. L.; Truhlar, D. G. Practical Methods for Including Torsional Anharmonicity in Thermochemical Calculations on Complex Molecules: The Internal-Coordinate Multi-Structural Approximation. *Phys. Chem. Chem. Phys.* **2011**, *13*, 10885–10907.
- (73) Johnson, D.; Marston, G. The Gas-Phase Ozonolysis of Unsaturated Volatile Organic Compounds in the Troposphere. *Chem. Soc. Rev.* **2008**, *37*, 699–716.
- (74) Kroll, J. H.; Sahay, S. R.; Anderson, J. G.; Demerjian, K. L.; Donahue, N. M. Mechanism of HO_x Formation in the Gas-Phase Ozone-Alkene Reaction. 2. Prompt versus Thermal Dissociation of Carbonyl Oxides to Form OH. *J. Phys. Chem. A* **2001**, *105*, 4446–4457.
- (75) Harrison, R. M.; Yin, J.; Tilling, R. M.; Cai, X.; Seakins, P. W.; Hopkins, J. R.; Lansley, D. L.; Lewis, A. C.; Hunter, M. C.; Heard, D. E.; et al. Measurement and Modelling of Air Pollution and Atmospheric Chemistry in the U.K. West Midlands Conurbation: Overview of the PUMA Consortium Project. *Sci. Total Environ.* **2006**, *360*, 5–25.
- (76) Yin, C.; Takahashi, K. How Does Substitution Affect the Unimolecular Reaction Rates of Criegee Intermediates? *Phys. Chem. Chem. Phys.* **2017**, *19*, 12075–12084.
- (77) Kuwata, K. T.; Luu, L.; Weberg, A. B.; Huang, K.; Parsons, A. J.; Peebles, L. A.; Rackstraw, N. B.; Kim, M. J. Quantum Chemical and Statistical Rate Theory Studies of the Vinyl Hydroperoxides Formed in trans-2-Butene and 2,3-Dimethyl-2-butene Ozonolysis. *J. Phys. Chem. A* **2018**, *122*, 2485–2502.
- (78) Huang, H.-L.; Chao, W.; Lin, J. J.-M. Kinetics of a Criegee Intermediate That Would Survive High Humidity and May Oxidize Atmospheric SO₂. *Proc. Natl. Acad. Sci. U. S. A.* **2015**, *112*, 10857–10862.
- (79) Kumar, M.; Busch, D. H.; Subramaniam, B.; Thompson, W. H. Barrierless Tautomerization of Criegee Intermediates via Acid Catalysis. *Phys. Chem. Chem. Phys.* **2014**, *16*, 22968–22973.
- (80) Berndt, T.; Voigtländer, J.; Stratmann, F.; Junninen, H.; Mauldin, R. L., III; Sipilä, M.; Kulmala, M.; Herrmann, H. Competing Atmospheric Reactions of CH₂OO with SO₂ and Water Vapour. *Phys. Chem. Chem. Phys.* **2014**, *16*, 19130–19136.
- (81) Chao, W.; Hsieh, J.-T.; Chang, C.-H.; Lin, J. J.-M. Direct Kinetic Measurement of the Reaction of the Simplest Criegee Intermediate with Water Vapor. *Science* **2015**, *347*, 751–754.
- (82) Vereecken, L.; Novelli, A.; Taraborrelli, D. Unimolecular Decay Strongly Limits the Atmospheric Impact of Criegee Intermediates. *Phys. Chem. Chem. Phys.* **2017**, *19*, 31599–31612.
- (83) Lee, Y.-P. Perspective: Spectroscopy and Kinetics of Small Gaseous Criegee Intermediates. *J. Chem. Phys.* **2015**, *143*, 020901.
- (84) Lewis, T. R.; Blitz, M. A.; Heard, D. E.; Seakins, P. W. Direct Evidence for a Substantive Reaction between the Criegee Intermediate, CH₂OO, and the Water Vapour Dimer. *Phys. Chem. Chem. Phys.* **2015**, *17*, 4859–4863.
- (85) Smith, M. C.; Chang, C.-H.; Chao, W.; Lin, L.-C.; Takahashi, K.; Boering, K. A.; Lin, J. J.-M. Strong Negative Temperature Dependence of the Simplest Criegee Intermediate CH₂OO Reaction with Water Dimer. *J. Phys. Chem. Lett.* **2015**, *6*, 2708–2713.
- (86) Taatjes, C. A.; Welz, O.; Eskola, A. J.; Savee, J. D.; Scheer, A. M.; Shallcross, D. E.; Rotavera, B.; Lee, E. P. F.; Dyke, J. M.; Mok, D. K. W.; et al. Direct Measurements of Conformer-Dependent Reactivity of the Criegee Intermediate CH₃CHOO. *Science* **2013**, *340*, 177–180.
- (87) Chhantyal-Pun, R.; Davey, A.; Shallcross, D. E.; Percival, C. J.; Orr-Ewing, A. J. A Kinetic Study of the CH₂OO Criegee Intermediate Self-Reaction, Reaction with SO₂ and Unimolecular Reaction Using

Cavity Ring-Down Spectroscopy. *Phys. Chem. Chem. Phys.* **2015**, *17*, 3617–3626.

(88) Buras, Z. J.; Elsamra, R. M. I.; Green, W. H. Direct Determination of the Simplest Criegee Intermediate (CH_2OO) Self Reaction Rate. *J. Phys. Chem. Lett.* **2014**, *5*, 2224–2228.

(89) Su, Y.-T.; Lin, H.-Y.; Putikam, R.; Matsui, H.; Lin, M. C.; Lee, Y.-P. Extremely Rapid Self-Reaction of the Simplest Criegee Intermediate CH_2OO and Its Implications in Atmospheric Chemistry. *Nat. Chem.* **2014**, *6*, 477–483.

(90) Womack, C. C.; Martin-Drumel, M.-A.; Brown, G. G.; Field, R. W.; McCarthy, M. C. Observation of the Simplest Criegee Intermediate CH_2OO in the Gas-Phase Ozonolysis of Ethylene. *Sci. Adv.* **2015**, *1*, No. e1400105.

(91) Fang, Y.; Liu, F.; Barber, V. P.; Klippenstein, S. J.; McCoy, A. B.; Lester, M. I. Deep Tunneling in the Unimolecular Decay of CH_3CHOO Criegee Intermediates to OH Radical Products. *J. Chem. Phys.* **2016**, *145*, 234308.

(92) Chhantyal-Pun, R.; Welz, O.; Savee, J. D.; Eskola, A. J.; Lee, E. P. F.; Blacker, L.; Hill, H. R.; Ashcroft, M.; Khan, M. A. H.; Lloyd-Jones, G. C.; et al. Direct Measurements of Unimolecular and Bimolecular Reaction Kinetics of the Criegee Intermediate $(\text{CH}_3)_2\text{COO}$. *J. Phys. Chem. A* **2017**, *121*, 4–15.

(93) Smith, M. C.; Chao, W.; Takahashi, K.; Boering, K. A.; Lin, J. J.-M. Unimolecular Decomposition Rate of the Criegee Intermediate $(\text{CH}_3)_2\text{COO}$ Measured Directly with UV Absorption Spectroscopy. *J. Phys. Chem. A* **2016**, *120*, 4789–4798.

(94) Long, B.; Bao, J. L.; Truhlar, D. G. Unimolecular Reaction of Acetone Oxide and Its Reaction with Water in the Atmosphere. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115*, 6135–6140.

(95) Skodje, R. T.; Truhlar, D. G.; Garrett, B. C. Vibrationally Adiabatic Models for Reactive Tunneling. *J. Chem. Phys.* **1982**, *77*, 5955–5976.

(96) Tautermann, C. S.; Voegelé, A. F.; Loerting, T.; Kohl, I.; Hallbrucker, A.; Mayer, E.; Liedl, K. R. Towards the Experimental Decomposition Rate of Carbonic Acid (H_2CO_3) in Aqueous Solution. *Chem. - Eur. J.* **2002**, *8*, 66–73.

(97) Skodje, R. T.; Truhlar, D. G.; Garrett, B. C. A General Small-Curvature Approximation for Transition-State-Theory Transmission Coefficients. *J. Phys. Chem.* **1981**, *85*, 3019–3023.

(98) Garrett, B. C.; Joseph, T.; Truong, T. N.; Truhlar, D. G. Application of the Large-Curvature Tunneling Approximation to Polyatomic Molecules: Abstraction of H or D by Methyl Radical. *Chem. Phys.* **1989**, *136*, 271–293.

(99) Fernandez-Ramos, A.; Truhlar, D. G. Improved Algorithm for Corner-Cutting Tunneling Calculations. *J. Chem. Phys.* **2001**, *114*, 1491–1496.

(100) Meana-Pañeda, R.; Truhlar, D. G.; Fernández-Ramos, A. Direct Dynamics Implementation of the Least-Action Tunneling Transmission Coefficient. Application to the $\text{CH}_4/\text{CD}_3\text{H}/\text{CD}_4 + \text{CF}_3$ Abstraction Reactions. *J. Chem. Theory Comput.* **2010**, *6*, 3015–3025.

(101) Marcus, R. A.; Coltrin, M. E. A New Tunneling Path for Reactions Such as $\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$. *J. Chem. Phys.* **1977**, *67*, 2609–2613.

(102) Bondi, D. K.; Connor, J. N. L.; Garrett, B. C.; Truhlar, D. G. Test of Variational Transition State Theory with a Large Curvature Tunneling Approximation against Accurate Quantal Reaction Probabilities and Rate Coefficients for Three Collinear Reactions with Large Reaction Path Curvature: $\text{Cl} + \text{HCl}$, $\text{Cl} + \text{DCl}$, and $\text{Cl} + \text{MuCl}$. *J. Chem. Phys.* **1983**, *78*, 5981–5989.

(103) Fernandez-Ramos, A.; Zgierski, M. Z. Theoretical Study of the Rate Constants and Kinetic Isotope Effects of the 1,2-Hydrogen-Atom Shift of Methoxyl and Benzyloxyl Radicals Assisted by Water. *J. Phys. Chem. A* **2002**, *106*, 10578–10583.

(104) Liu, Y. P.; Lynch, G. C.; Truong, T. N.; Lu, D. H.; Truhlar, D. G.; Garrett, B. C. Molecular Modeling of the Kinetic Isotope Effect for the [1,5]-Sigmatropic Rearrangement of cis-1,3-Pentadiene. *J. Am. Chem. Soc.* **1993**, *115*, 2408–2415.

(105) Lu, D.-H.; Truong, T. N.; Melissas, V. S.; Lynch, G. C.; Liu, Y.-P.; Garrett, B. C.; Steckler, R.; Isaacson, A. D.; Rai, S. N.; Hancock,

G. C.; et al. POLYRATE 4: A New Version of a Computer Program for the Calculation of Chemical Reaction Rates for Polyatomics. *Comput. Phys. Commun.* **1992**, *71*, 235–262.

(106) Garrett, B. C.; Truhlar, D. G. A Least-Action Variational Method for Calculating Multidimensional Tunneling Probabilities for Chemical Reactions. *J. Chem. Phys.* **1983**, *79*, 4931–4938.

(107) Garrett, B. C.; Abusalbi, N.; Kouri, D. J.; Truhlar, D. G. Test of Variational Transition State Theory and the Least-Action Approximation for Multidimensional Tunneling Probabilities against Accurate Quantal Rate Constants for a Collinear Reaction Involving Tunneling into an Excited State. *J. Chem. Phys.* **1985**, *83*, 2252–2258.

(108) Kreevoy, M. M.; Ostovic, D.; Truhlar, D. G.; Garrett, B. C. Phenomenological Manifestations of Large-Curvature Tunneling in Hydride-Transfer Reactions. *J. Phys. Chem.* **1986**, *90*, 3766–3774.

(109) Richardson, J. O. Microcanonical and Thermal Instanton Rate Theory for Chemical Reactions at All Temperatures. *Faraday Discuss.* **2016**, *195*, 49–67.

(110) Laude, G.; Calderini, D.; Tew, D. P.; Richardson, J. O. *Ab Initio* Instanton Rate Theory Made Efficient Using Gaussian Process Regression. *Faraday Discuss.* **2018**, *212*, 237–258.